

SPECTRAL ENTROPY AND SHARP STOCHASTIC WIDTHS OF COMPACT GROUP EXPERIMENTS

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ABSTRACT. We study finite stochastic registers whose transition rows may depend freely on the clock and horizon, with numerical decoders constrained to a compact convex output set. An entropy–transport estimate for mass-weighted conditional-centroid directions removes the logarithmic loss in the spectral-gap converse for alphabets with a physical identity letter. If the least unit-orbit dimension is p_* , every fixed subcritical error permits only $O(k^{2/p_*})$ cuts of width at most k , regardless of intervening widths. A quadratic orbit-hull construction gives the matching law when the target orbit has dimension p_* . In particular, legal $d \times d$ density-matrix outputs have width $\Theta(N^{d-1})$ at fixed positive subcritical error. For three explicit rational Bloch rotations, a classical Lubotzky–Phillips–Sarnak spectral bound makes the linear noisy law fully effective, whereas the exact cut profile is 5^t and persists under error at most $25^{-N}/16$. The earlier rational-unitary experiment, rational sparse-table compiler and adaptive classical observation construction are retained with their distinct hypotheses. These are label-width laws, not uniform workspace bounds; spectral constants and leading asymptotic constants are distinguished.

The resource counts persistent hidden labels, including unused padding. The horizon, clock, real table descriptions, table construction, exact arithmetic and exact sampling are uncharged. A numerical terminal decoder is not a sampled answer register; a process error is one budget for the whole control–output transcript.

1. INTRODUCTION AND REALIZATION MODEL

A finite stochastic register can encode a continuum of probability rows. Counting orbit points therefore gives a converse only at special extreme-output endpoints. At a fixed positive error, the relevant constraint is instead the repeated loss of conditional-centroid mass. We prove a sharp occupation bound by comparing that loss with the spectral dissipation of an auxiliary entropy.

Definition 1.1 (Clocked numerical realization). Let H be a compact group, $\mathcal{A}$ a finite nonempty command alphabet, and $F_{s,j} : H \rightarrow Y_j$ continuous interfaces for finite seed and query sets. Each Y_j is a compact convex set in a finite-dimensional normed space. At horizon N choose finite nonempty registers $S_0, \dots, S_N$, initialization rows E_s , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal decoder maps $D_j : S_N \rightarrow Y_j$. The realized output on $w = a_1 \cdots a_N$ is $E_s T_{1,a_1} \cdots T_{N,a_N} D_j$, and the physical product is $u_w = u_{a_N} \cdots u_{a_1}$. Write $W_{N,\varepsilon}$ for the least $\max_t |S_t|$ with worst-word, worst-seed and worst-query error at most ε .

Every advertised persistent label, including unused padding and retained private information, is counted. The horizon, external clock, arbitrary real tables, their construction and lookup, exact arithmetic and exact atomic sampling are uncharged in this primary invariant. A sampled answer register is a separate resource. A numerical density-matrix output is not quantum-state preparation. The rational compiler prices finite descriptions and fair-bit sampling separately.

Let R be a fixed-point-free orthogonal representation of a compact connected Lie group. Its least unit-orbit dimension p_* is the optimal exponent of a cap bound uniform in every direction,

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including normalized conditional centroids outside the target orbit. Under an L^2 spectral gap and a physical identity command, Theorem 4.5 gives

$$\#\{1 \leq t \leq N : |S_t| \leq k\} \leq C_\varepsilon k^{2/p_*}, \quad W_{N,\varepsilon} \geq c_\varepsilon N^{p_*/2}.$$

There is no restriction on intervening widths. The entropy potential is carried only along the narrow cuts, so there is no logarithmic mixing cost at each cut. A quadratic inner approximation of the target orbit hull supplies a positive common-row upper machine of width $O(N^{q/2})$, where q is that orbit's real dimension. When $p_* = q$, these are matching power laws. For legal density matrices of size d , they give $\Theta(N^{d-1})$ at every fixed subcritical tolerance, with strict amplitude or with positive error at unit amplitude.

The exact boundary is very different. One fixed rational unitary pair and pure seed has profile $2 \cdot 3^t - 1$ at cut t , still necessary for $\varepsilon \leq 625^{-N}/16$, but width $\Theta(N)$ at every fixed $0 < \varepsilon < 1/\sqrt{2}$. A second experiment, given by three explicit rational Bloch rotations and their inverses, has exact profile 5^t , stable for $\varepsilon \leq 25^{-N}/16$. The classical Lubotzky–Phillips–Sarnak theorem supplies the exact spherical spectral norm $\sqrt{5}/3$ for this second alphabet. Theorem 10.2 therefore gives the fully numerical bounds

$$\frac{(1 - \sqrt{2}\varepsilon)N}{6500} \leq W_{N,\varepsilon} \leq 52 + 104N/\varepsilon.$$

The constants are not optimal. For the first rational unitary pair, the constant imported from Bourgain–Gamburd remains non-effective. Neither freeness nor a spectral gap is inferred from finite enumeration.

The chain of new estimates is

$$\text{centroid-mass loss} \implies \text{transport of directions} \implies \text{entropy budget at narrow cuts}.$$

The kernel, transport inequality and entropy dissipation are proved below. Entropy–transport continuity and arithmetic spectral gaps are credited to their classical antecedents. The earlier mixing-block proof is retained as an alternative weaker bound, not used to hide a logarithm in the sharp result.

A prescribed joint terminal law is one possible query, but this model alone does not generate an observation process. A separate final section constructs causal instruments for classical repeatable probes under adaptive control. The structural companion proves clock removal under eventual approximate returns and classifies those repeatable-observation processes. Neither companion nor any predecessor PDF is a proof dependency of this quantitative article.

2. CONDITIONAL CENTROIDS AT ARBITRARY CUTS

Lemma 2.1 (Conditional-centroid contraction). *Let Z be a bounded random vector in a finite-dimensional Euclidean space and S the current hidden register. Choose an independent external word w whose physical action is orthogonal R_w . Suppose, for all unit vectors $u, v_1, \dots, v_k$,*

$$\mathbb{E}_w \max_{i \leq k} \langle v_i, R_w u \rangle \leq 1 - a, \quad 0 < a < 1.$$

If the endpoint register S' has at most k labels, then

$$\mathbb{E} \|\mathbb{E}[R_w Z \mid S']\| \leq (1 - a) \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

No restriction is imposed on intermediate widths. A fixed physically identity word cannot increase the latter conditional norm.

Proof. Let $z_s = \mathbb{E}[Z \mid S = s]$ and $T_w(s, i)$ be the actual composite stochastic row. The Markov property of the retained label gives

$$\Pr(S' = i) \mathbb{E}[R_w Z \mid S' = i] = \sum_s \Pr(S = s) \mathbb{E}_w T_w(s, i) R_w z_s.$$

Pair with a unit vector in each nonzero endpoint centroid's direction, sum in i , and use stochasticity to bound this by $\sum_s \Pr(S = s) \mathbb{E}_w \max_i \langle v_i, R_w z_s \rangle$. The hypothesized inequality gives the result. Zero masses and zero centroids cause no difficulty. For a fixed physical identity word the physical vector is unchanged, while the endpoint is a stochastic image of the incoming label; conditional Jensen proves nonincrease. Hidden rows on such a word need not be identity rows. $\square$

3. A UNIFORM ORBIT CRITERION FOR QUANTITATIVE WIDTH

The spherical estimate has an extension in which the dimension of the ambient feature space is not the relevant exponent. Two geometric quantities enter: a uniform small-ball exponent for every possible conditional-centroid direction, and the covering dimension of the particular target orbit. They need not agree. Keeping this distinction is essential for matrix-valued experiments.

Let G be a compact connected Lie group with normalized Haar measure m_G and a bi-invariant Riemannian distance. Let $R : G \rightarrow O(E)$ be a continuous orthogonal representation on a finite-dimensional real Euclidean space. Fix $z_0 \in E$ of norm one, put

$$\mathcal{O} = R(G)z_0, \quad Q = \text{conv}(\mathcal{O}), \quad q = \dim \mathcal{O},$$

and assume that $\mathcal{O}$ spans E and that E has no nonzero fixed vector. Thus $q \geq 1$. A finite alphabet in G contains the identity and carries a probability law μ for which

$$(3.1) \quad \|P^n f\|_2 \leq \lambda^n \|f\|_2 \quad \left(\int_G f dm_G = 0 \right), \quad Pf(g) = \mathbb{E}_{a \sim \mu} f(ua), \quad 0 < \lambda < 1.$$

Letters outside the support of μ are allowed. All command products and stochastic rows retain the conventions of Definition 1.1.

The uniform geometric hypothesis is that, for some $p > 0$, $A < \infty$ and $h_0 > 0$,

$$(3.2) \quad \sup_{u, v \in S(E)} m_G \{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

It is imposed on all unit vectors u , not only on $\mathcal{O}$. A conditional mean of points in $\mathcal{O}$, after normalization, need not belong to $\mathcal{O}$.

Lemma 3.1 (From orbit small balls to executable contraction). *Under (3.1)–(3.2), constants $c, C > 0$ and integers $B_k \leq C \log(k+2)$ can be chosen so that*

$$(3.3) \quad \mathbb{E}_{w \sim \mu^{B_k}} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - ck^{-2/p}$$

for every $u, v_1, \dots, v_k \in S(E)$. The constant c can be decreased so that the defect lies in $(0, 1)$ for every $k \geq 1$.

Proof. Choose $h = c_0 k^{-1/p} \leq \min(h_0, 1)$ with $Ac_0^p \leq 1/2$. Outside the union of the k open balls in (3.2),

$$1 - \max_i \langle v_i, R(g)u \rangle \geq h^2/2.$$

The union has measure at most $1/2$. Consequently the Haar mean of $f(g) = \max_i \langle v_i, R(g)u \rangle$ is at most $1 - a_k$, where $a_k = c_0^2 k^{-2/p}/4$. Repeated centers are allowed. All such functions are bounded by one in absolute value and have a common Lipschitz constant L : differentiating the finite-dimensional representation bounds the derivative uniformly on unit vectors.

Write $D = \dim G$, distinct from the cap exponent p and orbit dimension q . The ball calculation uses the fixed bi-invariant Riemannian metric on G . In exponential coordinates at the identity, Haar measure has a smooth positive density. Hence a metric ball of radius $\delta \leq \delta_0$ has measure at least $c_G \delta^D$. Average $P^B f$ over this ball. Left translations are isometries, so $P^B f$ has Lipschitz constant at most L . Cauchy–Schwarz and (3.1) give

$$|P^B f(1) - \bar{f}| \leq L\delta + 2c_G^{-1/2} \delta^{-D/2} \lambda^B.$$

Take δ to be a sufficiently small fixed multiple of a_k , also bounded by δ_0 . Then choose $B_k \geq 1$ to make the second term at most $a_k/4$. The first term is at most $a_k/4$ as well. Since $\log(a_k^{-1}) = O(\log(k+2))$, this requires $B_k \leq C \log(k+2)$. Products under μ^{B_k} are genuine words of one common length. Their reverse chronological order has the same law because the letters are independent with the same law. Thus $P^{B_k}f(1)$ is the expectation in (3.3). This proves the lemma with defect $a_k/2$. $\square$

Lemma 3.2 (Quadratic inner approximation of an orbit hull). *There are constants $C_1, C_2, \delta_0 > 0$ such that, for $0 < \delta \leq \delta_0$, points $z_1, \dots, z_K \in \mathcal{O}$, including z_0 , can be chosen with*

$$(3.4) \quad K \leq C_1 \delta^{-q}, \quad (1 - C_2 \delta^2)Q \subseteq \text{conv}\{z_1, \dots, z_K\} \subseteq Q, \quad 0 < 1 - C_2 \delta^2 < 1.$$

Proof. The orbit is a compact smooth embedded homogeneous manifold. Finitely many coordinate charts with bounded derivatives and inverse derivatives give an intrinsic δ -net of size at most $C\delta^{-q}$; adjoining z_0 changes only C . Use the induced Riemannian metric on the orbit. If the functional $x \mapsto \langle b, x \rangle$ attains its maximum at $x_b \in \mathcal{O}$, its derivative along the orbit vanishes there. The second derivative along unit-speed orbit geodesics has absolute value at most $C\|b\|$, uniformly in their starting points and directions, by compactness of the unit tangent bundle and smoothness of the embedding. A net point within intrinsic distance δ of x_b therefore satisfies

$$\langle b, z_i \rangle \geq h_Q(b) - C\|b\|\delta^2, \quad h_Q(b) = \max_{x \in Q} \langle b, x \rangle.$$

Haar averaging gives $\int R(g)z_0 dm_G = 0$. Moreover 0 lies in the interior of Q in E . Otherwise a nonzero supporting functional would be nonnegative on the orbit and have Haar mean zero; continuity and full Haar support would make it vanish on the orbit, contrary to its spanning E . Thus $r\mathbb{B}_E \subset Q$ for some $r > 0$, and $h_Q(b) \geq r\|b\|$. The support inequality becomes $\max_i \langle b, z_i \rangle \geq (1 - C\delta^2/r)h_Q(b)$. The support-function criterion for inclusion of compact convex sets proves (3.4). Decrease δ_0 to obtain the last bounds. $\square$

Theorem 3.3 (Geometric width and occupation). *Assume (3.1)–(3.2). Take one seed and one query with target $F(g) = \rho R(g)z_0$, output set Q , and Euclidean error. For $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$, put $\zeta = \rho - \varepsilon \in (0, 1]$. Every error- ε realization satisfies*

$$(3.5) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \log(k+2) \left(1 + k^{2/p} \log \frac{1}{\zeta}\right),$$

where C depends on the geometric and spectral-gap data but not on ρ, ε, N, k or the realization. In particular, at fixed parameters,

$$(3.6) \quad W_{N,\varepsilon} \geq c \left(\frac{N}{\log(N+2)} \right)^{p/2}.$$

For $0 < \rho < 1$ there is an exact common-row realization with

$$(3.7) \quad W_{N,0} \leq C \left(1 + \frac{N+1}{-\log \rho} \right)^{q/2}.$$

For $\rho = 1$ and fixed $0 < \varepsilon < 1$, the upper order is $O_\varepsilon((N+1)^{q/2})$. When $p = q$, the lower and upper exponents agree, with the displayed logarithmic loss in the lower bound. The constrained error threshold is exactly ρ , attained by the constant zero decoder.

Proof. Put $Z = R(u_w)z_0$ and let $D(S) \in Q \subseteq \mathbb{B}_E$ be the decoder. For each deterministic word, the error bound gives $\langle \mathbb{E}[D(S) | w], Z \rangle \geq \rho - \varepsilon$. Under any external word law, conditioning on the register yields

$$\zeta \leq \mathbb{E} \langle D(S), Z \rangle \leq \mathbb{E} \|\mathbb{E}[Z | S]\|.$$

The conditional-centroid Lemma 2.1 applies to the law of Lemma 3.1. The uniformity over every direction in (3.2) is precisely what justifies this application. It contracts the conditional norm by

$1 - a$ at a selected endpoint of width at most k , where $a = ck^{-2/p} \in (0, 1)$. The initial norm is one. Fixed identity-letter fillers cannot increase it, regardless of their rows or intervening widths. After J selected endpoints,

$$J[-\log(1 - a)] \leq \log(1/\zeta).$$

Greedily select narrow cuts at separation at least B_k , omitting the first $B_k - 1$ cuts. Each selected endpoint accounts for at most B_k cuts. The count is at most $B_k(1 + a^{-1} \log(1/\zeta))$, proving (3.5), also when $\zeta = 1$. In that case even one selected endpoint is impossible. More precisely one may take the integer bound

$$L_k = B_k \left(1 + \left\lceil c^{-1} k^{2/p} \log(1/\zeta) \right\rceil \right).$$

For a full width bound k , either $k \geq N^{p/2}$, or $\log(k + 2) \leq C_p \log(N + 2)$. In the latter case the occupation inequality gives $N \leq C_\zeta \log(N + 2) k^{2/p}$. These alternatives prove (3.6).

For the upper bound take the net in Lemma 3.2, and write $r = 1 - C_2 \delta^2$. The orbit hull is invariant under every command. Hence $rR(u_a)z_i$ is in the convex hull of the net points. For each of the finitely many command-vertex pairs, choose barycentric coordinates independently; no measurable selection is involved. Use them as row i of a stochastic matrix T_a , the same at every epoch within this horizon. Initialize at z_0 . The conditional row means then evolve as $r^t R(u_w)z_0$. Decode label i by $(\rho/r^N)z_i$, which belongs to Q if $r^N \geq \rho$, since $0 \in Q$. Choosing

$$\delta^2 = \min \left(\delta_0^2, \frac{1}{2C_2}, \frac{-\log \rho}{2C_2(N + 1)} \right)$$

gives this inequality by $\log(1 - x) \geq -2x$ for $0 \leq x \leq 1/2$. The net size gives (3.7). At unit amplitude and positive error use the exact construction for amplitude $1 - \varepsilon/2$; its discrepancy from the desired target is $\varepsilon/2$.

Finally, Haar mean zero and $\|R(g)z_0\| = 1$ imply

$$\int_G \|\rho R(g)z_0 - y\|^2 dm_G = \rho^2 + \|y\|^2.$$

Every constrained center therefore has worst error at least ρ , while $y = 0$ attains it. Below it (3.6) rules out bounded width. None of the constructions requires transition matrices to satisfy the physical group relations, and extra finitely many commands do not change the upper construction. $\square$

The geometric ingredients in Lemma 3.2 are the usual local geometry of compact homogeneous spaces; related covering estimates are developed in [15]. The additional realization statement is the common stochastic-row construction and its combination with a contraction uniform over conditional-centroid directions. A small-ball estimate restricted to the target orbit alone does not imply Theorem 3.3.

Remark 3.4 (Dependence of constants). The cap and executable-gap constants depend only on the fixed group with its bi-invariant metric, the representation, the uniform cap data (A, p, h_0) , and the declared L^2 gap λ . The hull constants also depend on the target orbit, its second derivatives, and the inradius of Q in its span E . The occupation bound further depends on $\zeta = \rho - \varepsilon$. None depends on the horizon, hidden transition rows, or masses of individual labels. The orbit is the smooth embedded quotient G/G_{z_0} and q is its real dimension. The logarithmic factor is the loss of this mixing-block argument. Theorem 4.5 removes it by a different entropy-transport argument; all statements and proofs of the block estimate remain valid.

4. SPECTRAL ENTROPY AND SHARP OCCUPATION BOUNDS

The pointwise mixing block in Lemma 3.1 costs a logarithm at each selected cut. We now replace that block argument by an entropy estimate along the selected cuts themselves. The auxiliary entropy is attached to a smoothed law of *centroid directions*; it is not the Shannon entropy of the hidden register. The normalization by centroid mass will be essential.

Let E have real dimension $m \geq 2$, let $\Sigma = \{u \in E : \|u\| = 1\}$, and let σ be normalized round measure on Σ . Write d_Σ for its geodesic distance and W_2 for the corresponding quadratic Wasserstein distance. All logarithms in this section are natural. Let $G \subseteq O(E)$ be compact, with normalized Haar measure m_G , and let λ be a probability on the command letters. On $L^2(\Sigma, \sigma)$ define

$$Pf(x) = \sum_a \lambda_a f(R(u_a)^{-1}x), \quad \Pi f(x) = \int_G f(R(g)^{-1}x) dm_G(g).$$

Here R denotes the given action; writing $G \subseteq O(E)$ identifies its image when convenient. The projection Π need not have one-dimensional range. Our spectral hypothesis is

$$(4.1) \quad \|Pf - \Pi f\|_2 \leq \kappa \|f - \Pi f\|_2, \quad 0 \leq \kappa < 1.$$

This is a bound on the whole function space, not merely on the finite-dimensional representation E . The group bound (3.1) implies (4.1), with the same κ : decompose the unitary representation on $L^2(\Sigma)$ into irreducibles; each nontrivial irreducible occurs in the regular representation, where the assumed bound holds (use the adjoint operator if needed for the inverse convention). The trivial summands are exactly the range of Π [13].

Remark 4.1 (The full spectral hypothesis). A contraction on E alone does not imply (4.1). For finitely many circle rotations, simultaneous rational approximation provides arbitrarily high characters on which the averaging eigenvalue is arbitrarily close to one, although its first harmonic may contract. Such an alphabet does not satisfy the full spectral hypothesis. This is why the present theorem does not replace the different Diophantine width laws by a spurious spherical exponent.

4.1. A compactly supported entropy potential. For $0 < h \leq 1$ set

$$(4.2) \quad K_h(u, x) = Z_h^{-1} \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2, \quad Z_h = \int_\Sigma \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2 d\sigma(x).$$

The normalizing constant is independent of u . If ν is a probability on Σ , write $K_h\nu(x) = \int K_h(u, x) d\nu(u)$. For a density f put

$$\mathcal{H}(f) = \int f \log f d\sigma, \quad \mathcal{I}(f) = \int_{\{f>0\}} \frac{\|\nabla f\|^2}{f} d\sigma, \quad 0 \log 0 = 0.$$

Lemma 4.2 (Entropy continuity for spherical mixtures). *There are constants $D_m \geq 1$ and $B_m < \infty$, depending only on m , such that every finitely supported probability ν satisfies*

$$(4.3) \quad 0 \leq \mathcal{H}(K_h\nu) \leq L_h := \log(D_m h^{-(m-1)}), \quad \mathcal{I}(K_h\nu) \leq B_m h^{-2}.$$

For any two finitely supported probabilities ν, η ,

$$(4.4) \quad |\mathcal{H}(K_h\nu) - \mathcal{H}(K_h\eta)| \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

For $m = 3$ one can take $D_3 = 12$ and $B_3 = 24$.

Proof. Put $s = \|x - u\|^2$. Polar coordinates give a density proportional to $s^{(m-3)/2}(1-s/4)^{(m-3)/2}$ on $0 < s < 4$. On $0 < s \leq 1$ its second factor is bounded above and below by positive dimension-dependent constants. Substitution $s = h^2 v$ therefore gives $Z_h \asymp_m h^{m-1}$. Also $\|\nabla s\|^2 = 4s - s^2$, so

$$\mathcal{I}(K_h(u, \cdot)) = \frac{4}{Z_h h^4} \int_{s < h^2} (4s - s^2) d\sigma \leq B_m h^{-2}.$$

Increasing D_m gives $K_h \nu \leq D_m h^{-(m-1)}$. Jensen's inequality gives the lower entropy bound, and the pointwise upper bound gives the upper one. Cauchy–Schwarz in a finite mixture shows that its Fisher information is at most the mixture of the individual Fisher informations.

Choose an optimal finite coupling (π_{ij}) of ν and η . Join its paired centers by shortest constant-speed spherical geodesics. Each path is $u_{ij}(t) = \exp(tA_{ij})u_i$, where A_{ij} is skew-symmetric, acts on a two-dimensional plane, and has operator norm $d_\Sigma(u_i, v_j)$. An antipodal pair is treated by choosing any such plane. Define

$$f_t = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), \cdot), \quad J_t(x) = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), x) A_{ij} x.$$

Rotation invariance gives $\partial_t f_t + \operatorname{div} J_t = 0$. Pointwise Cauchy–Schwarz and integration yield

$$\int_{\{f_t > 0\}} \frac{\|J_t\|^2}{f_t} d\sigma \leq \sum_{i,j} \pi_{ij} d_\Sigma(u_i, v_j)^2 = W_2(\nu, \eta)^2.$$

The preceding Fisher bound holds at every t . To justify entropy differentiation even where f_t vanishes, replace f_t by $(f_t + \delta)/(1 + \delta)$ and J_t by $J_t/(1 + \delta)$. The kernel is C^1 , the sphere is compact, and integration by parts followed by Cauchy–Schwarz gives

$$\left| \frac{d}{dt} \mathcal{H} \left(\frac{f_t + \delta}{1 + \delta} \right) \right| \leq \frac{1}{1 + \delta} \sqrt{\mathcal{I}(f_t)} \left(\int \frac{\|J_t\|^2}{f_t} d\sigma \right)^{1/2} \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

Here quotients at common zeros are zero; the displayed regularized identity can equivalently be obtained by smoothing the C^1 kernel. Integrate in t and let $\delta \downarrow 0$. Uniform boundedness and continuity of $x \log x$ at zero justify the limit and prove (4.4).

On $\Sigma = S^2$, the variable s has constant density $1/4$. Direct integration gives

$$Z_h = h^2/12, \quad \mathcal{I}(K_h(u, \cdot)) = 24/h^2 - 4.$$

These imply the asserted explicit constants. $\square$

The relation between entropy continuity and quadratic transport is classical; compare Polyanskiy–Wu [27, Proposition 1] for regular Euclidean densities. The proof above is included because its densities have compact support and zeros, and because the ambient sphere and the finite-mixture path, rather than Euclidean regularity, are used here.

Lemma 4.3 (Spectral entropy loss on a thin support). *Suppose (4.1) holds. Let f be a bounded probability density supported in a measurable set $A \subseteq \Sigma$ for which*

$$\Pi \mathbf{1}_A \leq \theta < 1.$$

Then

$$(4.5) \quad \mathcal{H}(f) - \mathcal{H}(Pf) \geq (1 - \kappa^2)(1 - \theta).$$

Proof. Orbital Cauchy–Schwarz gives $(\Pi \sqrt{f})^2 \leq (\Pi \mathbf{1}_A)(\Pi f)$ and hence $\|\Pi \sqrt{f}\|_2^2 \leq \theta$. The invariant and orthogonal components are preserved by P . Since $\|\sqrt{f}\|_2 = 1$,

$$\|P\sqrt{f}\|_2^2 \leq \|\Pi \sqrt{f}\|_2^2 + \kappa^2 \|(I - \Pi)\sqrt{f}\|_2^2 \leq 1 - (1 - \kappa^2)(1 - \theta).$$

For nonnegative numbers z_a with $z = \sum_a \lambda_a z_a$, the relative-entropy/Hellinger inequality gives

$$\sum_a \lambda_a z_a \log z_a - z \log z \geq \sum_a \lambda_a (\sqrt{z_a} - \sqrt{z})^2.$$

For completeness this follows by summing $t \log(t/s) - t + s \geq (\sqrt{t} - \sqrt{s})^2$; continuity handles zeros. Apply it pointwise to $z_a = f(R(u_a)^{-1}x)$ and integrate. Invariance of σ and Cauchy–Schwarz give

$$\mathcal{H}(f) - \mathcal{H}(Pf) \geq 2 \left(1 - \int \sqrt{Pf} P \sqrt{f} d\sigma \right) \geq 2(1 - \|P\sqrt{f}\|_2).$$

The inequality $2(1 - \sqrt{1-d}) \geq d$ for $0 \leq d \leq 1$ completes the proof. No logarithmic Sobolev inequality or pointwise mixing estimate has been assumed. $\square$

4.2. Transport paid for by loss of centroid mass. For a unit physical random vector X and a finite register S , put

$$(4.6) \quad c_s = \mathbb{E}[X \mid S = s], \quad \alpha = \sum_s \Pr(S = s) \|c_s\|, \quad \nu = \frac{1}{\alpha} \sum_{c_s \neq 0} \Pr(S = s) \|c_s\| \delta_{c_s / \|c_s\|}.$$

Only positive-mass labels contribute. The directional law is defined when $\alpha > 0$; its weights are not, in general, the probabilities of the hidden labels.

Lemma 4.4 (Centroid-loss transport). *Apply an independent letter $a \sim \lambda$, followed by any fixed physical identity-product word, using the actual stochastic rows of the machine. Let α', ν' be the quantities at the endpoint and suppose $\alpha' \geq \zeta > 0$. If $\lambda * \nu$ denotes the mixture of the rotated laws $R(u_a)_* \nu$, then $\delta = \alpha - \alpha' \geq 0$ and*

$$(4.7) \quad W_2(\lambda * \nu, \nu')^2 \leq \frac{3\pi^2}{\zeta} \delta.$$

There is no bound on any intervening register.

Proof. Write $T_a(s, i)$ for the actual composite row, including the fixed filler. Conditional independence of the row from the unrecorded past is precisely the machine definition. Thus

$$p'_i c'_i = \sum_{s,a} p_s \lambda_a T_a(s, i) R(u_a) c_s.$$

Set $w_{sai} = p_s \lambda_a T_a(s, i) \|c_s\|$, $q_i = \sum_{s,a} w_{sai}$, and $m_i = p'_i \|c'_i\|$. For $m_i > 0$ put $v_i = c'_i / \|c'_i\|$; otherwise choose an arbitrary unit vector. With $u_{sa} = R(u_a) c_s / \|c_s\|$ for nonzero terms,

$$(4.8) \quad \sum_{s,a} w_{sai} \|u_{sa} - v_i\|^2 = 2(q_i - m_i).$$

In particular $m_i \leq q_i$, $\sum_i q_i = \alpha$, $\sum_i m_i = \alpha'$, and $\delta \geq 0$.

Normalize the flows by α . They couple $\lambda * \nu$ to $\hat{\nu} = \sum_i (q_i / \alpha) \delta_{v_i}$. Chordal and geodesic distance satisfy $d_\Sigma(u, v) \leq (\pi/2) \|u - v\|$, so this coupling has geodesic quadratic cost at most $\pi^2 \delta / (2\alpha)$. If $\delta > 0$, the excess masses give a probability χ with

$$\hat{\nu} = (\alpha' / \alpha) \nu' + (\delta / \alpha) \chi.$$

Keeping the common part and transporting the remaining part across the sphere costs at most $\pi^2 \delta / \alpha$. The triangle inequality for W_2 and $(a + b)^2 \leq 2a^2 + 2b^2$ give $W_2(\lambda * \nu, \nu')^2 \leq 3\pi^2 \delta / \alpha$. Since $\alpha \geq \alpha' \geq \zeta$, this proves the result. The case $\delta = 0$ follows from the same couplings without an excess measure. Zero centroids and zero-probability labels require no division. $\square$

4.3. The sharp lower law. Assume, for some $A \geq 1$, $p > 0$, and $h_0 > 0$, the all-direction cap bound

$$(4.9) \quad \sup_{u,v \in \Sigma} m_G\{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

This positive-power cap bound excludes every fixed unit vector. Hence $E^G = \{0\}$ and the Haar mean of each orbit is zero; in particular $0 \in \text{conv } R(G)z_0$. The constants are fixed with the representation. For $k \geq 1$ define

$$(4.10) \quad c_0 = \min\{1, h_0, (2A)^{-1/p}\}, \quad h_k = c_0 k^{-1/p}, \quad \gamma = (1 - \kappa^2)/2, \quad L_k^{\text{ent}} = \log(D_m h_k^{-(m-1)}).$$

Theorem 4.5 (Occupation without logarithmic loss). *Let the command alphabet contain a physical identity letter, and assume (4.1) and (4.9). Fix a unit seed z_0 , let $Q = \text{conv } R(G)z_0$, and suppose the target is $\rho R(u_w)z_0$, with $0 < \rho \leq 1$ and legal decoder values in Q . Error is Euclidean error in E . For $0 \leq \varepsilon < \rho$ write $\zeta = \rho - \varepsilon$. Every horizon- N realization, including one redesigned at that horizon, satisfies*

$$(4.11) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq \left\lceil 1 + \frac{2L_k^{\text{ent}}}{\gamma} + \frac{3\pi^2 B_m}{\gamma^2 h_k^2} \frac{1 - \zeta}{\zeta} \right\rceil.$$

In particular, for fixed subcritical parameters there are constants $C_\zeta, c_\zeta > 0$ such that

$$(4.12) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C_\zeta k^{2/p}, \quad W_{N,\varepsilon} \geq c_\zeta N^{p/2}.$$

The intervening widths are unrestricted. The constants depend only on $m, A, p, h_0, \kappa, \zeta$ and the fixed round normalization.

Proof. If there are $J \geq 2$ positive cuts of width at most k , list them as $t_1 < \dots < t_J$. Use a fixed prefix up to t_1 . In each subsequent gap put one independent letter with law λ , then identity letters until the next selected cut. Finish by a fixed suffix. This is an external distribution on actual words of length N , chosen from the advertised width profile, not from the machine's private state. It does not require the machine's rows on an identity letter to be identities.

Let X be the current unit physical vector. Write α_j, ν_j for (4.6) at t_j . Wordwise correctness and $\|D_i\| \leq 1$ imply at the terminal cut

$$\zeta \leq \mathbb{E}\langle X_N, D(S_N) \rangle = \sum_i \Pr(S_N = i) \langle \mathbb{E}[X_N \mid S_N = i], D_i \rangle \leq \alpha_N.$$

The first inequality averages the conditional *mean* error on each deterministic word. It makes no assertion about a single sampled label's error. Conditional Jensen, or the flow calculation above, shows that centroid mass cannot increase along any gap or fixed suffix. Thus $1 \geq \alpha_j \geq \zeta$, and

$$\delta_j := \alpha_j - \alpha_{j+1} \geq 0, \quad \sum_{j < J} \delta_j \leq 1 - \zeta.$$

Fix $h = h_k$ and put $f_j = K_h \nu_j$. Each f_j is supported on at most k open chordal caps of radius h . The cap assumption gives $\Pi \mathbf{1}_{\{f_j > 0\}} \leq Akh^p \leq 1/2$. Lemma 4.3 therefore gives $\mathcal{H}(Pf_j) \leq \mathcal{H}(f_j) - \gamma$. Equivariance of the kernel says $Pf_j = K_h(\lambda * \nu_j)$. Lemmas 4.2 and 4.4 imply

$$\mathcal{H}(f_{j+1}) \leq \mathcal{H}(f_j) - \gamma + \frac{\pi\sqrt{3B_m}}{h\sqrt{\zeta}} \sqrt{\delta_j}.$$

Writing $M = J - 1$, summing, using $0 \leq \mathcal{H}(f_j) \leq L_h$, and applying Cauchy–Schwarz only over these M gaps, gives

$$(4.13) \quad \gamma M \leq L_h + \frac{\pi\sqrt{3B_m}}{h} \sqrt{M \frac{1 - \zeta}{\zeta}}.$$

Young's inequality bounds the second term by $\gamma M/2 + 3\pi^2 B_m(1 - \zeta)/(2\gamma h^2 \zeta)$. This proves (4.11); $J \leq 1$ is immediate. The logarithm in $L_h = O(1 + \log k)$ is additive, not multiplied by $k^{2/p}$. Since $1 + \log k = O_p(k^{2/p})$, the first bound in (4.12) follows. If the whole profile has width at most k , its N positive cuts are all counted, proving the second. $\square$

Remark 4.6 (Endpoint information and synchronization). The unrelaxed inequality (4.13) also holds. When $\zeta = 1$, it gives $J \leq 1 + L_k^{\text{ent}}/\gamma$, consistent with exponential exact width for extreme outputs. For fixed positive error it gives the polynomial law, but does not classify every joint limit $N \rightarrow \infty, \varepsilon \downarrow 0$.

If the identity letter is absent but physical identity-product words exist at every length at least g , greedily retain narrow cuts at spacing at least $g + 1$. Each retained cut accounts for at most $g + 1$ original narrow cuts. One independent command followed by a return of the remaining length fits each retained gap. The prefix and suffix may be any fixed words of their required lengths. The same proof multiplies the occupation bound by at most $g + 1$. Approximate returns with accuracy-dependent thresholds are not included in this quantitative extension without additional estimates.

Corollary 4.7 (Matching power laws). *In the setting of Theorem 3.3, replace its converse by (4.12). In particular, if the all-direction exponent p equals the real dimension q of the target orbit, then*

$$W_{N,\varepsilon} = \Theta((N + 1)^{q/2})$$

for every fixed $0 < \rho < 1$, $0 \leq \varepsilon < \rho$, and also for $\rho = 1$, $0 < \varepsilon < 1$. For a general fixed-point-free compact connected Lie representation one may take $p = p_*$ from Theorem 5.1; the lower and upper exponents are then $p_*/2$ and $q/2$, respectively, with no logarithmic loss in the lower bound.

Proof. Apply Theorem 4.5. The upper common-row construction in Theorem 3.3 is exact for $\rho < 1$ and has $O((N + 1)^{q/2})$ labels. At $\rho = 1$ and positive error use its amplitude-slack version. The two bounds match when $p = q$. Theorem 5.1 supplies the remaining assertion. $\square$

The endpoint $\rho = 1, \varepsilon = 0$ is not an assertion of the matching polynomial corollary. No exact leading constant or asymptotic ratio is claimed. The spectral constant is effective only when a certified value is supplied; a fully numerical example is given below.

5. THE LEAST ORBIT DIMENSION AND UNIFORM SMALL BALLS

The geometric hypothesis needed for a stochastic converse concerns every conditional-centroid direction, not only the prescribed target orbit. For a fixed finite-dimensional orthogonal representation of a compact connected Lie group, this hypothesis follows from a rank invariant. Throughout this section Haar measure is normalized, and the Lie algebra has a fixed invariant inner product. No constants uniform in the dimension or in the representation are asserted.

Theorem 5.1 (Uniform caps from the least orbit dimension). *Let G be a compact connected Lie group and $R : G \rightarrow O(E)$ a continuous finite-dimensional representation with $E^G = \{0\}$ and $E \neq \{0\}$. Write $D = \dim G$ and choose a basis $X_1, \dots, X_D$ of its Lie algebra, using the same symbols for their infinitesimal actions on E . Set*

$$(5.1) \quad p_* = \min_{\|u\|=1} \text{rank}[X_1 u \cdots X_D u] = \min_{\|u\|=1} \dim R(G)u.$$

Then $p_* \geq 1$, and there is $C < \infty$ such that

$$(5.2) \quad \sup_{\|u\|=1, v \in E} \mu\{g : \|R(g)u - v\| < h\} \leq Ch^{p_*} \quad (0 < h \leq 1).$$

The exponent is optimal as a uniform exponent: it cannot be replaced by any larger exponent with a constant independent of u, v, h .

Proof. The tangent space to the orbit at u is the span of the $X_i u$. An orbit of dimension zero is a singleton, since G is connected; its point would be fixed. This proves $p_* \geq 1$. Put $r = p_*$ and fix orthonormal coordinates on E .

For each unit u , some r -row, r -column minor of $[X_1 u \cdots X_D u]$ is nonzero. The maximum of the absolute values of these finitely many minors is a continuous positive function on the unit sphere. Its minimum is therefore positive. The entries are uniformly bounded. Consequently there is $\sigma > 0$ such that for every unit u one can choose r input indices and r output coordinates whose derivative matrix A_u has least singular value at least σ .

For each of the finitely many permutations of the chosen Lie algebra basis consider the product chart

$$\phi(x, y) = \exp(x_1 X_{i_1}) \cdots \exp(x_r X_{i_r}) \exp(y_1 X_{i_{r+1}}) \cdots \exp(y_{D-r} X_{i_D}).$$

All these charts are diffeomorphisms on one sufficiently small cube about zero. Their images contain a common identity neighborhood W , and their Haar densities are bounded above by a common constant. By shrinking the cube once more, uniform continuity of the derivatives on the unit sphere and the finite list of charts ensures the following. For every unit u , choose the minor above and let $\pi_u : E \rightarrow \mathbb{R}^r$ be its coordinate projection. For all (x, y) in the cube,

$$(5.3) \quad \|\partial_x(\pi_u R(\phi(x, y))u) - A_u\| \leq \sigma/2.$$

For fixed y , the map $f_{u,y}(x) = \pi_u R(\phi(x, y))u$ is injective. Indeed integration of the derivative on the segment between x and x' gives

$$\|f_{u,y}(x) - f_{u,y}(x')\| \geq \|A_u(x - x')\| - \frac{1}{2}\sigma\|x - x'\| \geq \frac{1}{2}\sigma\|x - x'\|.$$

Every singular value of its derivative is at least $\sigma/2$. The change-of-variables formula therefore bounds the r -dimensional volume of the inverse image of a ball of radius h by $\omega_r(2/\sigma)^r h^r$. A ball in E projects into a ball of the same radius in these coordinates. Integrating in y and using the bounded Haar density proves the required estimate on this chart, uniformly in u and in the center v .

Choose finitely many right translates Wg_j covering G . On such a translate apply the preceding argument to the unit vector $R(g_j)u$. For its selected minor, use the corresponding permuted chart; that chart contains W . Haar invariance and a finite sum give (5.2) for sufficiently small h . Enlarging C covers the remaining $h \leq 1$. This proof does not require a spectral decomposition of the arbitrary center v .

Finally choose u_* whose orbit has dimension r . The orbit is the smooth embedded compact homogeneous space G/G_{u_*} . The pushforward of Haar measure is its invariant probability measure, with positive smooth density in local coordinates. At u_* , sufficiently small ambient chordal balls have measure bounded below by ch^r for some $c > 0$. Thus no larger uniform exponent is possible. $\square$

Proposition 5.2 (An algebraic rank certificate). *Suppose the infinitesimal matrices X_i are supplied with exact real algebraic entries, and are known to specify the representation in Theorem 5.1. Then p_* is computable by real quantifier elimination. A positive algebraic lower bound for the largest $p_* \times p_*$ minor on the unit sphere is computable as well. If a unit seed z_0 is algebraic, its orbit dimension q is the rank of $[X_1 z_0 \cdots X_D z_0]$.*

Proof. For each integer r ask whether a unit vector exists on which all $(r+1) \times (r+1)$ minors vanish. These are finitely many polynomial conditions over real algebraic numbers; the first affirmative answer is p_* . With $r = p_*$, the sum of squares of all r -minors has a positive minimum on the unit sphere, by the first paragraph of the preceding proof. Quantifier elimination computes that minimum as a real algebraic number with isolating data. If there are M minors, their largest absolute value is at least $\sqrt{\gamma/M}$, where γ is that minimum. Algebraic rank at the supplied seed gives the last assertion. $\square$

The proposition computes the rank invariant from *supplied* infinitesimal matrices. It neither certifies that arbitrary input matrices integrate to a compact group nor computes a random-walk spectral gap.

Corollary 5.3 (Uniform width exponents without a separate cap hypothesis). *In the orbit realization model of Theorems 3.3 and 4.5, assume the declared spectral gap, but replace its small-ball hypothesis by $E^G = \{0\}$. Let p_* be (5.1), and let q be the real dimension of the unit target orbit spanning E . For fixed $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$,*

$$W_{N,\varepsilon} \geq cN^{p_*/2}.$$

For $0 < \rho < 1$ the exact upper bound is $C(N+1)^{q/2}$; for $\rho = 1$ it holds at each fixed positive error. Every fixed width has the occupation bound of Theorems 3.3 and 4.5. Error is measured in the Euclidean norm defining the legal orbit hull.

Proof. Apply Theorem 5.1 to obtain its uniform cap estimate with exponent p_* . The lower and upper arguments of Theorems 3.3 and 4.5 then apply without alteration. $\square$

The group dimension D , least orbit dimension p_* , and target orbit dimension q need not coincide. For the standard sphere representation $p_* = q = d-1$. For conjugation on traceless Hermitian $d \times d$ matrices, an orbit with eigenvalue multiplicities $m_1, \dots, m_b$ has dimension $d^2 - \sum_i m_i^2$. A nonzero traceless matrix has $b \geq 2$, and the largest possible sum is $(d-1)^2 + 1$. Thus $p_* = 2(d-1)$, attained by a centered rank-one projection. The explicit Grassmannian estimates below also provide this exponent; they are retained as a direct geometric proof with stated dimension-dependent constants.

The converse now uses Theorem 4.5, rather than paying the pointwise-mixing block length at each narrow cut. Thus no logarithm separates the bounds when $p_* = q$. A supplied algebraic infinitesimal representation makes p_* computable; this statement does not compute a spectral gap or assert dimension-uniform constants.

6. POSITIVE MATRIX REALIZATIONS IN EVERY DIMENSION

Let $d \geq 2$ and let $\mathcal{H}_d^0$ be the real Euclidean space of traceless Hermitian $d \times d$ matrices, with scalar product $\langle A, B \rangle = \text{tr}(AB)$ and Frobenius norm $\|\cdot\|_F$. Let

$$\mathcal{D}_d = \{D = D^* \succeq 0 : \text{tr } D = 1\}, \quad c_d = I/d, \quad r_d = \sqrt{1 - 1/d}.$$

This section concerns numerical outputs in $\mathcal{D}_d$. The requirement that every decoder value be positive with trace one is part of the model; it is not imposed merely on the final average.

Lemma 6.1 (Small balls on the complex Grassmannian). *Let P be a fixed rank- m orthogonal projection in $\mathbb{C}^d$, with $1 \leq m < d$. For Haar $U \in \text{SU}(d)$ there is $C_{d,m} < \infty$ such that*

$$\Pr\{\|UPU^* - Q\|_F < h\} \leq C_{d,m} h^{2m(d-m)} \quad (0 < h \leq 1)$$

for every rank- m projection Q . For rank one, writing $\text{dist}_{\text{pr}}(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, one has $\text{dist}_{\text{pr}}(P, Q) = \|P - Q\|_F / \sqrt{2}$ and exactly

$$(6.1) \quad \Pr\{\text{dist}_{\text{pr}}(UPU^*, Q) < h\} = h^{2(d-1)} \quad (0 \leq h \leq 1).$$

Proof. Conjugation is transitive, so fix $Q = \text{diag}(I_m, 0)$. A projection sufficiently close to Q has range the graph of a unique matrix $Z \in \mathbb{C}^{(d-m) \times m}$, with

$$P(Z) = \begin{pmatrix} I \\ Z \end{pmatrix} (I + Z^* Z)^{-1} \begin{pmatrix} I & Z^* \end{pmatrix}.$$

This notation means the product of the three displayed matrices. The map and its inverse are smooth near zero, and its derivative at zero is $Z \mapsto \begin{pmatrix} 0 & Z^* \\ Z & 0 \end{pmatrix}$, of Frobenius norm $\sqrt{2}\|Z\|_F$. The invariant probability measure is normalized Riemannian volume on this compact homogeneous

manifold; in this chart its density is smooth and bounded on a sufficiently small closed neighborhood. The inverse chart puts a Frobenius h -ball inside a Euclidean Ch -ball in real dimension $2m(d - m)$. This proves the bound for small h ; increasing the constant covers the remaining interval. Transitivity makes the constant independent of Q .

For rank one choose $Q = e_1 e_1^*$. A uniform unit vector can be obtained by normalizing independent standard complex Gaussian coordinates. If X is the squared modulus of the first coordinate before normalization and Y is the sum of the other squared moduli, then X is exponential of mean one and Y is an independent sum of $d - 1$ such variables. For $0 < z < 1$,

$$\Pr \left\{ \frac{X}{X + Y} > 1 - z \right\} = \mathbb{E} \exp \left(-\frac{1 - z}{z} Y \right) = z^{d-1}.$$

Put $z = h^2$. The endpoint cases follow by continuity. This proves (6.1), and also shows that dist_{pr} is the Frobenius chordal distance divided by $\sqrt{2}$. $\square$

Lemma 6.2 (A uniform conjugacy-orbit exponent). *There is $C_d < \infty$ such that*

$$(6.2) \quad \sup_{\substack{A, B \in \mathcal{H}_d^0 \\ \|A\|_F = \|B\|_F = 1}} m_G \{U : \|UAU^* - B\|_F < h\} \leq C_d h^{2(d-1)} \quad (0 < h \leq 1).$$

The exponent cannot be increased uniformly over A and B .

Proof. Write the eigenvalues of A as $a_1 \geq \dots \geq a_d$. Since their sum is zero and their squared sum is one, $a_1 - a_d \geq d^{-1/2}$. Some adjacent gap therefore satisfies

$$(6.3) \quad a_m - a_{m+1} \geq \gamma_d := \frac{1}{\sqrt{d}(d-1)}.$$

Let P_A be the spectral projection onto the first m eigenvectors. It is well defined despite possible multiplicities within either block.

Two conjugates A', A'' of A , with eigenbases x_i, y_j in this order, obey the exact identity

$$(6.4) \quad \|A' - A''\|_F^2 = \sum_{i,j} (a_i - a_j)^2 |\langle x_i, y_j \rangle|^2.$$

Expanding both traces proves it. The cross-block terms and (6.3) imply

$$(6.5) \quad \|P_{A'} - P_{A''}\|_F \leq \gamma_d^{-1} \|A' - A''\|_F,$$

since the squared norm on the left is exactly the sum of the two cross-block overlap sums.

If the set in (6.2) is nonempty, choose one member U_0 . Every other member has $\|UAU^* - U_0 A U_0^*\|_F < 2h$. Thus its rank- m spectral projection lies within $2h/\gamma_d$ of $U_0 P_A U_0^*$. The Haar law of $U P_A U^*$ is the invariant Grassmannian law. Lemma 6.1 bounds the measure by $C_{d,m} (2h/\gamma_d)^{2m(d-m)}$ for $h \leq \gamma_d/2$. Since $m(d-m) \geq d-1$, and there are finitely many m , this gives (6.2) with a constant independent of all eigenvalue multiplicities. Larger h are covered by increasing C_d .

For sharpness take $A = B = (P - c_d)/r_d$ with P rank one. By (6.1), for sufficiently small h the probability is

$$(r_d^2 h^2 / 2)^{d-1}.$$

It cannot be bounded by a constant times $h^{p'}$ as $h \downarrow 0$ for any $p' > 2(d-1)$. $\square$

The spectral split, rather than an assumption that all centroids remain rank one, supplies the uniformity needed by the stochastic contraction. The conjugation representation has ambient dimension $d^2 - 1$, but its worst small-ball exponent is $2(d-1)$.

Theorem 6.3 (Unitary matrix-orbit width). *Let a finite alphabet in $\text{SU}(d)$ contain the identity and support a probability law satisfying (3.1). Fix a rank-one projection P_0 . For $0 < a \leq 1$, set*

$$(6.6) \quad F_a(U) = c_d + a(UP_0U^* - c_d), \quad Y = \mathcal{D}_d,$$

with Frobenius error. Its exact bounded-width threshold is ar_d , attained by the one-label decoder c_d . For $0 \leq \varepsilon < ar_d$, put $\zeta = a - \varepsilon/r_d$. Every realization satisfies

$$(6.7) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \left(1 + \log(k+1) + \frac{1-\zeta}{\zeta} k^{1/(d-1)} \right).$$

For fixed $0 < a < 1$ and $0 \leq \varepsilon < ar_d$,

$$(6.8) \quad cN^{d-1} \leq W_{N,\varepsilon} \leq C(N+1)^{d-1},$$

and the upper realization can be exact. The same orders hold for $a = 1$ and every fixed $0 < \varepsilon < r_d$.

More explicitly, for $a < 1$ and $\varepsilon = 0$ set $a' = a$; for $\varepsilon > 0$ set $a' = a - \varepsilon/(2r_d)$. Whenever $a' < 1$, an error at most $\varepsilon/2$ realization has

$$(6.9) \quad K \leq C_d \left(1 + \frac{N+1}{-\log a'} \right)^{d-1}$$

labels. In the zero-error case this realization is exact. In (6.7) the constant depends only on d and the spectral-gap data, not on the amplitude or tolerance. Extra finitely many unitary commands are allowed in both bounds.

Proof. Center at c_d and divide all outputs by r_d . The resulting legal set is

$$Q = \text{conv}\{(P - c_d)/r_d : P \text{ rank one}\} \subseteq \mathbb{B}_{\mathcal{H}_d^0}.$$

The last inclusion follows from $\text{tr } D^2 \leq 1$ for $D \in \mathcal{D}_d$. The orbit spans $\mathcal{H}_d^0$: if a traceless Hermitian A has $\text{tr}(AP) = 0$ for every rank-one P , then $v^*Av = 0$ for every v , hence $A = 0$. Conjugation-fixed matrices are scalar, so there are no fixed vectors in this traceless space. The orbit is complex projective space, of real dimension $2(d-1)$. Lemma 6.2 and Theorem 4.5, with $p = q = 2(d-1)$, prove the lower and occupation bounds, as well as the radius statement after multiplying the norm by r_d .

We give a direct inner approximation to make positivity of the upper rows transparent. Choose a maximal δ -separated family of rank-one projections P_i for dist_{pr} , containing P_0 . Its $\delta/2$ balls are disjoint, so (6.1) gives

$$K \leq (2/\delta)^{2(d-1)}.$$

For any traceless Hermitian A , put $M = \lambda_{\max}(A)$. When $A \neq 0$, $M > 0$ and $\lambda_{\min}(A) \geq -(d-1)M$. Approximate a top eigenline by P_i at distance at most δ . Spectral decomposition then gives

$$(6.10) \quad \text{tr}(AP_i) \geq M - (M - \lambda_{\min}(A))\delta^2 \geq (1 - d\delta^2)M.$$

The trivial $A = 0$ case is automatic. Comparing support functions in $\mathcal{H}_d^0$, with $K_0 = \mathcal{D}_d - c_d$, yields

$$(1 - d\delta^2)K_0 \subseteq \text{conv}\{P_i - c_d\} \subseteq K_0.$$

Put $r = 1 - d\delta^2 > 0$. For each actual command U_b choose one row of barycentric coordinates satisfying

$$(6.11) \quad \sum_j T_b(i, j)(P_j - c_d) = rU_b(P_i - c_d)U_b^*.$$

Rows are nonnegative and sum to one by construction. Initialize at P_0 and decode label i as

$$D_i = c_d + (a'/r^N)(P_i - c_d).$$

This is in $\mathcal{D}_d$ when $r^N \geq a'$: it is a convex combination of c_d and P_i . Choosing

$$\delta^2 = \min \left(\frac{1}{4d}, \frac{-\log a'}{2d(N+1)} \right)$$

ensures this inequality. Induction in (6.11) gives mean output exactly $F_{a'}(u_w)$. The target discrepancy is $|a - a'|r_d = \varepsilon/2$ when $\varepsilon > 0$. The cardinal bound proves (6.9). The case $a = 1, \varepsilon = 0$ is deliberately excluded from this dilation construction and is treated exactly in the next section. $\square$

The family contains the qubit sphere as $d = 2$, but for $d \geq 3$ the pure-state orbit is a proper curved subset of the unit sphere in $\mathcal{H}_d^0$. Thus (6.8) does not follow by substituting the ambient dimension into the spherical theorem. The entropy argument removes the logarithmic loss. No optimal leading constant, exact asymptotic ratio, or dimension-uniform constant is claimed.

Corollary 6.4 (Algebraic generating alphabets). *Theorem 6.3 applies to the identity together with any finite symmetric algebraic-entry set generating a dense subgroup of $\mathrm{SU}(d)$, equipped with a lazy symmetric law. Such a finite alphabet exists for every $d \geq 2$.*

Proof. The spectral gap is the imported theorem of Bourgain–Gamburd [16]. Laziness makes its one-sided gap an absolute L^2 contraction on mean-zero functions. Existence can be seen by placing two algebraic irrational-angle $\mathrm{SU}(2)$ rotations in every adjacent pair of coordinate directions. For example, use the quaternion lifts in the explicit adjacent-block construction below within each pair. Each pair generates its full $\mathrm{SU}(2)$ subgroup. Their Lie algebras contain the adjacent off-diagonal matrix units and their brackets, which generate $\mathfrak{su}(d)$. The connected compact closure is consequently $\mathrm{SU}(d)$. All entries remain algebraic. The spectral-gap constant is not computed by the finite regression checks. $\square$

6.1. A finite whole-law encoding. Let $q_0 = d^2 - 1$ and choose a Frobenius-orthonormal basis $H_1, \dots, H_{q_0}$ of $\mathcal{H}_d^0$. Since $\|H_l\|_{\mathrm{op}} \leq 1$, the $2q_0$ matrices

$$E_{l,\pm} = \frac{I \pm H_l/2}{2q_0}$$

are positive and sum to I . Put $\Phi(D)_{l,\pm} = \mathrm{tr}(E_{l,\pm}D)$ and $Y_\Phi = \Phi(\mathcal{D}_d)$. Every entry is at least $1/(4q_0)$. The map is injective, because

$$(6.12) \quad D - c_d = \sum_l 2q_0(\Phi(D)_{l,+} - \Phi(D)_{l,-})H_l.$$

On the vector space parallel to the affine span of Y_Φ , use the norm of the reconstructed matrix on the right. Then Φ is an affine isometry, and replacing each legal matrix decoder by its image preserves every clocked or stationary width exactly. This is one whole categorical law with a compact convex legality constraint, not separate binary event tolerances. For ordinary total variation the explicit norm comparison is

$$\frac{\|D - D'\|_{\mathrm{F}}}{4q_0} \leq \mathrm{TV}(\Phi(D), \Phi(D')) \leq \frac{\|D - D'\|_{\mathrm{F}}}{4\sqrt{q_0}}.$$

It follows directly by comparing the ℓ^1 and ℓ^2 norms of the basis coefficients. Enlarging the legal output set from Y_Φ to the entire simplex is a different realization problem. In particular the extreme-output argument below cannot be transferred to that enlarged set: its target vectors are interior points of the simplex.

Remark 6.5 (An explicit algebraic alphabet in every dimension). In each adjacent coordinate plane $(j, j+1)$ embed the two matrices

$$\frac{1}{5} \begin{pmatrix} -3 + 4i & 0 \\ 0 & -3 - 4i \end{pmatrix}, \quad \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad 1 \leq j < d,$$

and act identically on all other coordinates. Include their inverses and the identity. Each pair has infinite-order eigenvalues, since its trace $-6/5$ is not an integer, and its two torus closures generate the adjacent $SU(2)$. The adjacent skew-Hermitian matrix units and their brackets span $\mathfrak{su}(d)$. The closed generated subgroup is therefore $SU(d)$. Bourgain–Gamburd is applied to this actual finite algebraic set, not merely to the existence of an unspecified set. The resulting spectral-gap constant is not numerically evaluated here.

The Frobenius chordal distance between rank-one projections is $\sqrt{2}$ times the projective distance used in the net construction. In the conjugacy-cap comparison, choosing one member of a nonempty target ball avoids imposing any spectral condition on its arbitrary center. All adjacent-gap and cap constants here depend on d . The amplitude $a' = a - \varepsilon/(2r_d)$ in the positive-error upper bound deliberately spends only half the available error.

7. EXTREME OUTPUTS AND THE EXACT ENDPOINT

The strict amplitude slack in the polytope construction is not a technical convenience that can simply be removed. At an extreme output, convex averaging has no room to hide a different physical state. This observation gives an exact finite-horizon formula, independently of spectral gap or synchronization.

Theorem 7.1 (The extreme-orbit formula). *Let a group act by affine bijections on a compact convex set Y . Let $x_0 \in Y$ have an orbit consisting of extreme points of Y , and let a finite nonempty command alphabet act through this group. There is one seed and one terminal query, with target $u_w x_0$. Put*

$$\mathcal{O}_t = \{u_w x_0 : |w| = t\}, \quad n_t = |\mathcal{O}_t|.$$

Every exact clocked stochastic realization at horizon N satisfies

$$(7.1) \quad |S_t| \geq n_t \quad (0 \leq t \leq N).$$

One deterministic realization attains all these inequalities simultaneously. Consequently

$$(7.2) \quad W_{N,0} = n_N.$$

Neither an identity letter nor an exact return is required.

Proof. Fix t and one suffix v of length $N - t$. Such a suffix exists because the alphabet is nonempty; at $t = N$ use the empty word. Let $D_v(i) \in Y$ be the conditional expected terminal decoder after this fixed suffix, when the cut- t label is i . Convexity guarantees that it lies in Y . For every prefix w of length t , exact correctness says

$$u_v u_w x_0 = \sum_i p_w(i) D_v(i),$$

where p_w is the actual distribution at that cut. Extremality forces $D_v(i) = u_v u_w x_0$ whenever $p_w(i) > 0$. Since the action of u_v is injective, prefixes with different physical outputs have disjoint nonempty supports in S_t . This proves (7.1), including the terminal cut.

Conversely, at cut t store the current point of $\mathcal{O}_t$. A command sends this point to its image in $\mathcal{O}_{t+1}$; the terminal decoder is the stored point. The initialization is deterministic. Finally, any one fixed command gives an injection $\mathcal{O}_t \rightarrow \mathcal{O}_{t+1}$, so n_t is nondecreasing. The peak of this realization is n_N , proving (7.2). $\square$

The same conclusion holds for an orbit on a Euclidean unit sphere with legal output set its convex hull: every orbit point is extreme by strict convexity of the ambient Euclidean ball. For density matrices, a rank-one projection is extreme. Indeed, if $P = \sum_i p_i D_i$ with $D_i \geq 0$ and $\text{tr } D_i = 1$, then $\text{tr}((I - P)D_i) = 0$ for every positive weight. Positivity forces D_i to be supported on the range of P , and hence $D_i = P$.

The cut argument fixes one suffix, whose affine action is injective. The cardinalities are nondecreasing because the nonempty alphabet contains a letter whose injective action maps each reachable set into the next one. No sampled-answer floor is included in this width.

8. A FIXED RATIONAL EXPERIMENT WITH AN EXPONENTIAL EXACT PROFILE

We now give fixed, explicit matrices and a rational pure seed. The freeness and trivial-stabilizer assertions have elementary finite-field proofs; no generic choice of generators or transcendental seed is used. The spectral-gap input is needed only for the noisy lower bound.

Lemma 8.1 (A quaternion free triple). *In the unit quaternions put*

$$a = (1 + 2\mathbf{i})/\sqrt{5}, \quad b = (1 + 2\mathbf{j})/\sqrt{5}, \quad c = (1 + 2\mathbf{k})/\sqrt{5}.$$

Every nonempty freely reduced word in $a^{\pm 1}, b^{\pm 1}, c^{\pm 1}$ is nonscalar. In particular these elements freely generate a group of rank three, also after passage to the projective unitary group.

Proof. Use the Hamilton relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$. Modulo five these relations are respected by

$$\mathbf{i} \mapsto \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{j} \mapsto \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}, \quad \mathbf{k} \mapsto \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

The six numerators $1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}$ therefore have respective images

$$(8.1) \quad \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \\ \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

They have rank one and their image lines are the six distinct points of $\mathbb{P}^1(\mathbb{F}_5)$. The kernel line of each matrix is the image line of the matrix with the opposite sign. Hence two consecutive matrices have zero product exactly when their letters are inverses. Writing each rank-one matrix as a column times a row shows that a product of them is nonzero whenever no adjacent pair is inverse.

A reduced word of length $n \geq 1$ is $5^{-n/2}Q$, where Q is an integral Hamilton quaternion of norm 5^n . Its image modulo five is nonzero by the preceding observation. Were Q scalar, its integer scalar entry t would satisfy $t^2 = 5^n$. For odd n this is impossible; for even $n \geq 2$ it forces $t = \pm 5^{n/2}$, making the image zero modulo five. This contradiction proves the assertion. Since the scalar unit quaternions are ± 1 , it proves both freeness statements. $\square$

Use the faithful complex representation

$$\rho(t + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \begin{pmatrix} t + xi & y + zi \\ -y + zi & t - xi \end{pmatrix}.$$

Define the following matrices, all of whose real and imaginary parts are rational:

$$(8.2) \quad U = \frac{1}{5} \begin{pmatrix} -3 + 4i & 0 \\ 0 & -3 - 4i \end{pmatrix}, \quad V = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad P_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Here $U = \rho(a^2)$, $V = \rho(b^2)$, and P_0 is a rank-one density matrix. The alphabet is exactly

$$(8.3) \quad \mathcal{A} = \{I, U, U^*, V, V^*\}.$$

Theorem 8.2 (Explicit rational exact-noisy separation). *For the fixed input (8.2)–(8.3), let the terminal target be $F(g) = gP_0g^*$ and require each decoder value to be a legal density matrix in $\mathcal{D}_2$. Error is Frobenius error. At every horizon $N \geq 0$ the exact minimum command-width profile is*

$$(8.4) \quad |S_t|_{\min} = 2 \cdot 3^t - 1 \quad (0 \leq t \leq N), \quad W_{N,0} = 2 \cdot 3^N - 1.$$

In fact the same profile is necessary, and is attained by the exact machine, throughout the closed range

$$(8.5) \quad 0 \leq \varepsilon \leq 625^{-N}/16.$$

For each fixed $0 < \varepsilon < 1/\sqrt{2}$, constants $c_\varepsilon, C_\varepsilon > 0$ satisfy

$$(8.6) \quad c_\varepsilon N \leq W_{N,\varepsilon} \leq C_\varepsilon(N+1).$$

At $\varepsilon \geq 1/\sqrt{2}$ one command label suffices. The upper bound in (8.6) has a uniform rational-table and fair-bit construction, specified in Theorem 9.2.

Proof. First, $\langle U, V \rangle$ is free of rank two: a nontrivial reduced word in $a^{\pm 2}, b^{\pm 2}$ expands to a nontrivial reduced word in $a^{\pm 1}, b^{\pm 1}$, and Lemma 8.1 applies. If a member g of this subgroup fixes P_0 , then it commutes with P_0 . It consequently commutes with

$$\rho(c) = (I + 2i(2P_0 - I))/\sqrt{5}.$$

A nonempty reduced word w in a, b cannot commute with c : wcw^{-1} is reduced, has length $2|w|+1$, and cannot equal c in the free group. Thus the projective stabilizer of P_0 in $\langle U, V \rangle$ is trivial.

The subgroup is dense in $SU(2)$. Both U and V have trace $-6/5$. If either had root-of-unity eigenvalues, its trace would be a rational algebraic integer and hence an integer, a contradiction. Its powers are therefore dense in its one-dimensional torus. The Lie algebra of the closed generated subgroup contains the $\mathbf{i}$ and $\mathbf{j}$ directions and their bracket, the $\mathbf{k}$ direction. It is all of $\mathfrak{su}(2)$; connectedness of $SU(2)$ implies density.

By freeness and the trivial stabilizer, the reachable pure states are in bijection with the reduced word ball of radius t . Identity padding makes this precisely the set reachable at length t . Its size is $1 + 4 \sum_{j=0}^{t-1} 3^j = 2 \cdot 3^t - 1$. The extreme-orbit profile theorem (Theorem 7.1) gives (8.4). Its proof uses one fixed suffix at each cut, not a query-dependent suffix.

We prove the stronger assertion (8.5) directly. Write $D = (I + x \cdot \sigma)/2$, with $x \in B_3$, for the Bloch coordinates of a density matrix. Then

$$(8.7) \quad \|D - D'\|_F = \|x - x'\|/\sqrt{2}.$$

The rotations induced by U, V, U^*, V^* have entries in $25^{-1}\mathbb{Z}$. At cut t , different reachable unit Bloch vectors have coordinates in $25^{-t}\mathbb{Z}$, and hence mutual Euclidean distance at least 25^{-t} . Fix any suffix of length $N - t$. Its action is an isometry, so the same separation holds for the corresponding terminal targets. For each cut label let $z_i \in B_3$ be its conditional terminal decoder mean after this fixed suffix. For a prefix with pure target n , error at most ε gives

$$\left\| \sum_i p_i z_i - n \right\| \leq \sqrt{2}\varepsilon, \quad \sum_i p_i \|z_i - n\|^2 \leq 2 \left(1 - \left\langle \sum_i p_i z_i, n \right\rangle \right) \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is therefore within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (8.5), twice this radius is at most $\sqrt{\sqrt{2}/2} 25^{-N} < 25^{-t}$. These closed balls are disjoint for different prefixes. They require distinct labels and prove the pointwise lower bound at every cut. Deterministic exact storage gives the simultaneous upper bounds.

For the positive-error converse, apply the spectral-gap theorem of Bourgain–Gamburd [16, Theorem 1] to the *displayed* algebraic generators U, V . The density just proved verifies its hypothesis. The symmetric walk becomes an absolute L^2 contraction off the constants after making it lazy, for example with mass $1/2$ at I and $1/8$ at each other letter. The legal density-matrix width theorem in dimension two gives the lower bound in (8.6). The rational construction proved next gives its upper bound without using a spectral gap. The center $I/2$ is at distance $1/\sqrt{2}$ from every pure state, which gives the final threshold assertion; the lower bound excludes bounded width at any smaller fixed error. $\square$

This is a finite-input separation, not merely an existence statement in an arbitrary-real advice model. Every length- N target has real and imaginary entries with denominator dividing $2 \cdot 25^N$, hence bit length $O(N + 1)$. Neither the exponential exact lower bound nor its robust extension uses a spectral-gap theorem. The spectral-gap constant in the noisy converse remains non-effective in the present proof; no numerical value is inferred from finite word tests. The logarithmic interval in (8.6) is stated explicitly rather than suppressed.

9. A UNIFORM RATIONAL QUBIT COMPILER

The preceding example also admits a constructive upper realization. The compiler below constructs its rows rather than assuming that an optimal realization has already been supplied. Its persistent labels and finite table resources are accounted for separately.

Lemma 9.1 (A rational spherical inner hull). *For $m \geq 2$ an integer, let $\mathcal{V}_m$ consist of the distinct points*

$$(9.1) \quad v_{j,\ell,\eta} = \eta \frac{(2jm, 2\ell m, m^2 - j^2 - \ell^2)}{m^2 + j^2 + \ell^2}, \quad -m \leq j, \ell \leq m, \quad \eta \in \{-1, 1\}.$$

Put $r = 1 - m^{-2}$ and add the vertex 0 as a distinguished label. Then

$$rB_3 \subseteq \text{conv}(\mathcal{V}_m) \subseteq B_3, \quad K := 1 + |\mathcal{V}_m| \leq 1 + 2(2m + 1)^2 \leq 13m^2.$$

The points have $O(\log(m+1))$ -bit rational coordinates. For any supplied rational rotation $A \in SO(3)$ and any label v , the point rAv has a rational barycentric expression using 0 and at most three members of $\mathcal{V}_m$. Such an expression can be found by finitely enumerating triples and solving rational 3×3 systems.

Proof. The stereographic parametrization $\psi(s, t) = (2s, 2t, 1 - s^2 - t^2)/(1 + s^2 + t^2)$ has differential norm at most two. Every point in the closed northern hemisphere has a preimage in the unit disk, hence in $[-1, 1]^2$. Rounding its coordinates to the nearest multiples of $1/m$ gives a point of (9.1) within distance $\sqrt{2}/m$. Negation treats the southern hemisphere. For every unit n there is thus $v \in \mathcal{V}_m$ with $\langle n, v \rangle = 1 - \|n - v\|^2/2 \geq 1 - m^{-2}$. The support-function criterion proves the inner inclusion. The stated cardinality bound follows since $m \geq 2$; duplicates are discarded.

The polytope contains a neighborhood of zero. The ray through a nonzero rAv reaches its boundary in a face. Triangulate a facet containing that face using its vertices. A triangle containing the ray endpoint, together with zero, gives the desired expression, allowing zero coefficients on lower-dimensional faces. Its three nonzero vertices are linearly independent: their affine plane is a supporting plane not through zero. Enumerate all such triples, solve for their three coefficients, and retain any solution with nonnegative coefficients and sum at most one. The remaining mass is assigned to zero. Cramer's rule gives rational coefficients; for $v = 0$ use the point mass at zero. All arithmetic is finite, and no selection theorem is needed. $\square$

Theorem 9.2 (Constructed finite-input upper realizations). *Let the alphabet consist of a fixed finite list of rational Bloch rotations, and let the seed be the first coordinate unit vector. These include the explicit experiment (8.2). For a rational $0 < \beta < 1$ and horizon $N \geq 0$, choose*

$$(9.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{N}{1 - \beta}} \right\rceil \right), \quad r = 1 - m^{-2}.$$

There is an exactly correct rational stochastic realization of the matrix target $F_\beta(g) = I/2 + \beta(gP_0g^ - I/2)$ on at most $13m^2$ command labels. Each command row has at most four nonzero entries and is independent of the command epoch within the chosen horizon.*

For any rational $0 < \varepsilon < 1/\sqrt{2}$, the pure target F_1 admits an error- ε realization on $O(1 + N/\varepsilon)$ labels with rational legal matrix decoders and dyadic rows. It can be sampled with at most Nb fair bits, where it suffices to take

$$(9.3) \quad b = \left\lceil \log_2 \frac{24(N+1)}{\varepsilon} \right\rceil.$$

Both constructions are uniform algorithms polynomial in N , $(1 - \beta)^{-1}$ or ε^{-1} , and the input bit lengths. This is not a polynomial bound in the binary length of N or in the logarithm of the inverse accuracy.

Proof. The net includes the first coordinate unit vector, so initialization is deterministic. For each letter A and each net label v_i , select the rational row in Lemma 9.1 whose barycentric mean is rAv_i . The zero label stays at zero. Induction shows that the mean Bloch vector of the label at time t is $r^t A_{a_t} \cdots A_{a_1} e_1$. At label v_i use the matrix decoder

$$(9.4) \quad D_i = \frac{1}{2} \left(I + \frac{\beta}{r^N} v_i \cdot \sigma \right).$$

Bernoulli's inequality and (9.2) give $r^N \geq 1 - N/m^2 \geq \beta$. Thus every decoder is positive semidefinite of trace one. Its expectation is exactly F_β , for every length- N word; the same rows work for all those words. Formula (9.4) also gives $I/2$ at the zero label.

For the pure target choose $\eta = \min(\varepsilon, 1/2)$ and $\beta = 1 - \eta$. The amplitude loss contributes Frobenius error $\eta/\sqrt{2} \leq \varepsilon/\sqrt{2}$. Keep the rational decoders and round each sparse probability row to denominator 2^b without enlarging its support: round down the first at most three masses and give the last mass the remainder. Its total-variation error is at most $3 \cdot 2^{-b}$. Stochastic contraction bounds the terminal distribution discrepancy by $3N2^{-b}$. The Frobenius diameter of $\mathcal{D}_2$ is $\sqrt{2}$; therefore the additional error is at most $3\sqrt{2}N2^{-b} \leq \varepsilon/4$ under (9.3). Since $1/\sqrt{2} + 1/4 < 1$, the total is strictly less than ε . One uniform b -bit integer samples each row. There is no initialization draw and no sampled quantum output in this count.

For completeness we give finite construction costs. Enumerating the net uses $O(m^2)$ rational points. For each of $|\mathcal{A}|K$ rows, enumerating at most $\binom{K-1}{3}$ triples and solving fixed-size systems uses polynomially many rational operations. Cramer's rule shows that the exact row coefficients have bit length $O(\log(m+1) + B_{\mathcal{A}})$, where $B_{\mathcal{A}}$ bounds the supplied rotation-entry bit lengths. The denominator r^N has bit length $O(N \log(m+1))$. Expanded decoders therefore have $O(B_\beta + (N+1) \log(m+1))$ bits each. A naive full enumeration needs $O(|\mathcal{A}|K^4)$ fixed-size rational solves, each of polynomial bit cost. This proves the algorithmic assertion as well as the bit bounds. $\square$

Sparse row tables require $O(|\mathcal{A}|K(B_{\mathcal{A}} + \log(K+1) + \log(m+1)))$ bits before rounding, and $O(|\mathcal{A}|K(b + \log(K+1)))$ bits after rounding. Expanded matrix decoders require $O(K(B_\beta + (N+1) \log(m+1)))$ bits. The persistent label uses $\lceil \log_2 K \rceil$ bits; the horizon counter, temporary sampler, and arithmetic working storage are separately charged. Unlike a statement only about output bit length, the preceding proof gives a terminating polynomial table-construction algorithm for this specific family. It does not solve general minimum-width realization or certify a spectral gap. The full legal density-matrix constraint is maintained at every rounding step.

Proposition 9.3 (Supplied rational unit seeds). *Theorem 9.2 remains valid for any supplied rational unit Bloch seed, with the label bound replaced by $13m^2 + 1$.*

Proof. Add the seed to the rational unit net if it is absent. The inner-ball inclusion is unchanged, every contracted rotated vertex still has an exact rational sparse row, and initialization is now deterministic at the added seed. The same induction and terminal rescaling apply. $\square$

The implementation accepts the supplied finite rotation list and unit seed as exact rational data. Orthogonality, determinant one and unit seed norm are checked by rational identities before table allocation.

10. A RATIONAL EXPERIMENT WITH EXPLICIT SPECTRAL CONSTANTS

The original rational pair in Theorem 8.2 now has a sharp linear fixed-error law, but its spectral constant is still supplied existentially by Bourgain–Gamburd. We give a second finite input for which every constant in a useful linear lower bound is numerical. The deep input here is the classical Lubotzky–Phillips–Sarnak theorem, not a new spectral-gap calculation.

Let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices in their usual order, and put $D(x) = (I + \sum_j x_j \sigma_j)/2$ for $x \in \mathbb{B}_3$. Thus $D(\mathbb{B}_3) = \mathcal{D}_2$ and $\|D(x) - D(y)\|_F = \|x - y\|/\sqrt{2}$. Consider the following *rational Bloch rotations*:

$$(10.1) \quad A_1 = \frac{1}{5} \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & -4 \\ 0 & 4 & -3 \end{pmatrix}, \quad A_2 = \frac{1}{5} \begin{pmatrix} -3 & 0 & 4 \\ 0 & 5 & 0 \\ -4 & 0 & -3 \end{pmatrix}, \quad A_3 = \frac{1}{5} \begin{pmatrix} -3 & -4 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Take the alphabet

$$\mathcal{A}_{\text{LPS}} = \{I, A_1, A_1^\top, A_2, A_2^\top, A_3, A_3^\top\}$$

and the rational pure seed e_1 , so $D(e_1) = P_0$ as in (8.2). The induced unitary lifts can be taken as $(I - 2i\sigma_j)/\sqrt{5}$. They are algebraic, not rational matrices; the rational input in (10.1) describes the physical Bloch action directly. This is a different alphabet from the rational unitary pair of Theorem 8.2.

Lemma 10.1 (The classical six-rotation spectral bound). *Let P_{LPS} average uniformly over the six nonidentity letters in $\mathcal{A}_{\text{LPS}}$. On normalized S^2 ,*

$$(10.2) \quad \|P_{\text{LPS}}\|_{L_0^2(S^2) \rightarrow L_0^2(S^2)} = \frac{\sqrt{5}}{3}.$$

Their generated subgroup is dense in $SO(3)$.

Proof. The matrices in (10.1) are the adjoint actions of the unit Hamilton quaternions $(1 + 2\mathbf{i})/\sqrt{5}, (1 + 2\mathbf{j})/\sqrt{5}, (1 + 2\mathbf{k})/\sqrt{5}$. Together with their inverses they are exactly the norm-five set in the Lubotzky–Phillips–Sarnak theorem [28, Theorems 1.3 and 1.5]; see the precise restatement [29, Theorem 1.1], which gives $2\sqrt{p}/(p+1)$ for this set at $p = 5$. This proves (10.2). The cited theorem uses the theory of automorphic forms and Deligne’s theorem; no finite matrix enumeration replaces it.

Density can also be checked directly. Each rotation has cosine $-3/5$. If its angle were a rational multiple of 2π , twice this cosine would be a rational algebraic integer, contrary to $-6/5 \notin \mathbb{Z}$. Its powers are dense in its circle subgroup. The three coordinate-axis circle subgroups generate $SO(3)$, as follows either from their infinitesimal generators and brackets or from Euler rotations. Hence the closed generated subgroup is $SO(3)$. $\square$

Theorem 10.2 (An effective linear noisy law and an exponential endpoint). *For the fixed input (10.1), the terminal target is $D(A_w e_1)$, every decoder belongs to $\mathcal{D}_2$, and error is Frobenius error. For $0 < \varepsilon < 1/\sqrt{2}$ put $\zeta = 1 - \sqrt{2}\varepsilon$. Every admissible realization satisfies*

$$(10.3) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq 1 + 6 \log(12k) + 648\pi^2 \frac{1 - \zeta}{\zeta} k \leq \frac{6500}{\zeta} k.$$

In particular,

$$(10.4) \quad \frac{\zeta N}{6500} \leq W_{N,\varepsilon} \leq 52 + \frac{104N}{\varepsilon}.$$

For rational ε in this interval the upper machine has uniformly constructed rational legal decoders and dyadic stochastic rows, each supported on at most four labels. It uses at most Nb fair bits for hidden transitions, where $b = \lceil \log_2(24(N+1)/\varepsilon) \rceil$.

At every horizon $N \geq 0$, the simultaneous exact minimum profile is

$$(10.5) \quad |S_t|_{\min} = 5^t \quad (0 \leq t \leq N).$$

The same profile remains necessary and is attained throughout

$$(10.6) \quad 0 \leq \varepsilon \leq 25^{-N}/16.$$

For $\varepsilon \geq 1/\sqrt{2}$, one label suffices.

Proof. For a spherical chordal cap of radius $h \leq 1$, normalized area is exactly $h^2/4$. Consequently the support of a mixture of k kernels K_h occupies at most $kh^2/4$ of each group orbit. Take $h = k^{-1/2}$. In Lemma 4.3 we may take

$$\theta = 1/4, \quad \kappa^2 = 5/9, \quad \gamma = (1 - \kappa^2)(1 - \theta) = 1/3.$$

Lemma 4.2 gives $L_h = \log(12k)$ and $B_3 = 24$. The proof of Theorem 4.5, with these constants and the normalization $\zeta = 1 - \sqrt{2}\varepsilon$, gives the first inequality in (10.3) as a real bound before taking an integer ceiling. For $k \geq 1$, $\log(12k) \leq 3k$, and $\pi^2 < 10$. Thus its right side is at most $19k + 6480(1 - \zeta)k/\zeta \leq 6500k/\zeta$. Counting all positive cuts proves the lower bound in (10.4).

For the upper bound apply the rational compiler of Theorem 9.2 to the displayed seven-letter list and the first-axis seed. At rational tolerance take $\beta = 1 - \varepsilon/2$ and $m = \max\{2, \lceil \sqrt{2N/\varepsilon} \rceil\}$. It gives an exact rational realization of $D(\beta A_w e_1)$ on at most $13m^2 \leq 52 + 52N/\varepsilon$ labels, with sparse common rows. The target change has Frobenius norm $\varepsilon/(2\sqrt{2})$. The stated dyadic precision adds at most $3\sqrt{2}N2^{-b} \leq \sqrt{2}\varepsilon/8$, so the total is less than ε . This proves the looser advertised numerical upper bound and the fair-bit statement. For a real tolerance choose a rational $\tilde{\varepsilon} \in (\varepsilon/2, \varepsilon)$ and use the same construction at $\tilde{\varepsilon}$; the bound $52 + 104N/\varepsilon$ follows. The algorithmic assertion uses rational input, not an oracle for an unencoded real number.

We next prove the exact profile independently of the spectral bound. Lemma 8.1 shows that the three norm-five quaternion generators freely generate modulo scalars, hence their adjoint rotations form a free group on a, b, c . The stabilizer of e_1 in this free group is exactly $\langle a \rangle$. Indeed, a rotation fixing e_1 commutes with the rotation a about that axis. In a free group the centralizer of this generator is its cyclic subgroup: write a reduced word outside $\langle a \rangle$ as $a^r v a^s$, where v is nonempty and begins and ends with neither a nor a^{-1} . Its conjugate of a reduces to $a^r v a v^{-1} a^{-r}$, which is not a . The converse, that every power of a fixes e_1 , is immediate.

The orbit is therefore indexed by the right cosets of $\langle a \rangle$. Each coset has a unique reduced representative with no terminal $a^{\pm 1}$, obtained by removing the maximal terminal power of a . It has least word length in that coset. At length $\ell \geq 1$ there are $4 \cdot 5^{\ell-1}$ such representatives. Identity padding gives exactly the ball of these cosets at command length t , whose cardinality is $1 + 4 \sum_{\ell=1}^t 5^{\ell-1} = 5^t$. Theorem 7.1 applies because pure states are extreme in $\mathcal{D}_2$, proving (10.5) at every cut; deterministic orbit storage attains the entire profile at once.

For robustness, distinct cut- t Bloch targets have coordinates in $5^{-t}\mathbb{Z}^3$ and hence distance at least 5^{-t} . One fixed suffix of length $N - t$ preserves these distances. If $z_i \in \mathbb{B}_3$ is the conditional terminal decoder mean from cut label i , wordwise error at a pure target n implies

$$\sum_i p_i \|z_i - n\|^2 \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (10.6), twice this radius is strictly less than $5^{-N} \leq 5^{-t}$. The corresponding closed balls for distinct prefix orbit points are disjoint, and so require distinct labels. The exact construction attains this bound. No fixed positive tolerance

is asserted to lie in (10.6) for all horizons. Finally $D(0) = I/2$ is at Frobenius distance $1/\sqrt{2}$ from every pure state, proving the one-label assertion. $\square$

Remark 10.3 (What is effective). The input, orbit counts, robust tolerance, spectral norm, constants in (10.3)–(10.4), and rational compiler are explicit. The constant 6500 is conservative; it is not an optimal leading constant. The spectral norm is imported from [28, 29], not inferred from finite harmonic tests. For the earlier five-letter rational-unitary example, Theorem 4.5 removes the logarithm but does not supply a numerical Bourgain–Gamburd constant. Neither statement claims a complete two-parameter phase diagram as the error tends to zero. The finite-table and fair-bit resources refer to a classical hidden label and a numerical matrix output, not quantum-state preparation.

11. A RATIONAL COMPILER FOR ADAPTIVE OBSERVATION PROCESSES

The common-row construction also yields a causal approximation, but its proof requires control of the full observation law. Correct terminal means alone are not enough. We give that additional argument here.

The physical state is $x \in B_3$, initialized at a supplied rational $x_0 \in S^2$. A finite rational rotation list $\{A_a\} \subset SO(3)$ acts by $x \mapsto A_a x$ and emits a null symbol. A finite probe list has rational affine probability laws

$$(11.1) \quad q_{j,y}(x) = b_{j,y} + \langle \ell_{j,y}, x \rangle, \quad \sum_y b_{j,y} = 1, \quad \sum_y \ell_{j,y} = 0, \quad b_{j,y} - \|\ell_{j,y}\| \geq \eta > 0.$$

A probe emits a fresh draw and leaves x unchanged. Controls may depend on the entire observed transcript. A simulator has finite retained labels and a nonnegative joint emission/transition matrix for each control and output; these matrices may depend on horizon and epoch. Let $\text{PW}_{N,\varepsilon}$ denote its minimum peak label count for one total-variation budget ε on the entire transcript, uniformly over all adaptive randomized policies. This paragraph defines a causal model independently of the terminal matrix-output model.

Put $A = \max_j \sum_y \|\ell_{j,y}\|^2$ and $C_0 = \max(3, \max_j (M_j - 1))$. The lower probability bound in (11.1) is used only for the relative-entropy upper estimate, not for the finite-testing converse below.

Theorem 11.1 (Adaptive rational and fair-bit realization). *For rational $0 < \varepsilon < 1$ and integer $N \geq 0$, choose*

$$(11.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{4AN^2}{\eta\varepsilon^2}} \right\rceil \right), \quad r = 1 - m^{-2}, \quad b = \max \left(0, \left\lceil \log_2 \frac{2C_0 \max(1, N)}{\varepsilon} \right\rceil \right).$$

There is an error- ε causal realization on at most $13m^2 + 1$ labels. Before probability rounding every rotation row is rational, has at most four nonzero entries, and has barycentric mean $rA_a v$. Probe emissions leave the label unchanged. After rounding, all emission and transition probabilities are dyadic, with at most b fair bits per control and Nb bits in total. The tables are stationary within the chosen horizon.

An exact rational algorithm constructs these tables in time polynomial in the numerical parameters $N, \varepsilon^{-1}, A/\eta$, the explicit input size, and the input bit lengths. In particular, for fixed probe and rotation lists,

$$(11.3) \quad \text{PW}_{N,\varepsilon} = O(1 + N^2/\varepsilon^2).$$

The bit count now includes the actual finite probe outputs as well as hidden transitions; it does not describe quantum-state preparation.

Proof. Take the rational net V_m from Lemma 9.1, including zero, and add x_0 if needed. This adds at most one label and cannot shrink the convex hull. Initialize deterministically at x_0 . For every rotation and label v , the same exact triple search supplies a row with mean rA_av . At probe j , emit y with probability $q_{j,y}(v)$ and retain v . All points lie in B_3 .

Fix an arbitrary policy. On the same finite trajectory space of controls, outputs and hidden labels, introduce an auxiliary law P : hidden labels follow the just constructed rotation kernels, but every probe emits $q_{j,y}(X)$ from the true physical state X , independently of the hidden label conditional on the public history. Its public marginal is exactly the target experiment. Let Q be the constructed simulator law. The policy kernels, initial label and hidden rotation kernels are the same under P, Q ; only probe emissions differ.

Under P , if the current physical and hidden points are X, V , both X and every nonzero hidden vertex are unit vectors. At a rotation step, for $V \neq 0$,

$$\begin{aligned} \mathbb{E}[\|V' - A_a X\|^2 \mid X, V, a] &\leq 2 - 2r\langle V, X \rangle \\ &= r\|V - X\|^2 + 2(1 - r). \end{aligned}$$

For $V = 0$ choose its row to remain at zero; the same upper bound holds because the left side is one and the right side is $2 - r \geq 1$. The added seed is a unit vector, so these two cases exhaust the state set. This is why a general interior seed has not been silently added to the theorem. A probe changes neither point. Iteration, with the control selected adaptively, gives

$$(11.4) \quad \mathbb{E}_P\|V_t - X_t\|^2 \leq 2t/m^2.$$

The inequality is unconditional; no posterior error bound after an unlikely observation is assumed.

For probability vectors p, q with $q_y \geq \eta$, the inequality $\log z \leq z - 1$ gives

$$D(p\|q) \leq \sum_y (p_y - q_y)^2 / q_y.$$

Consequently the conditional divergence at a probe is at most $A\|X - V\|^2/\eta$. The finite-trajectory relative-entropy chain rule cancels the common policy and rotation kernels. Applying (11.4) at the probe times gives

$$(11.5) \quad D(P\|Q) \leq \frac{2AN^2}{\eta m^2}, \quad \text{TV}(P_{\text{pub}}, Q_{\text{pub}}) \leq \sqrt{D(P\|Q)/2} \leq \frac{N}{m} \sqrt{A/\eta} \leq \varepsilon/2.$$

Here marginalization can only decrease divergence. These elementary entropy facts apply to the joint classical trajectory, not to a quantum state. One may verify the total-variation inequality by partitioning into the set where $P \geq Q$ and its complement: the log-sum inequality reduces divergence to a Bernoulli divergence, whose second derivative in its first parameter is at least four. This yields $D(P\|Q) \geq 2\text{TV}(P, Q)^2$. The chain rule follows by factoring each trajectory probability into its conditional kernels. All probe probabilities in Q are positive, so absolute continuity causes no exception. If $A = 0$, every probe is constant and the divergence is zero.

Round each rotation row and each probe row by rounding all but its last positive mass down to denominator 2^b and assigning the remainder to the last. A row supported on d points changes in total variation by at most $(d - 1)2^{-b}$. Thus every one-step joint kernel changes by at most $C_0 2^{-b}$. Coupling the two label-transcript processes until their first discrepancy proves, for every adaptive policy, an additional path error at most $NC_0 2^{-b} \leq \varepsilon/2$. Combining this with (11.5) proves the asserted error. Each rounded row can be sampled by one uniform b -bit integer and cumulative integer masses; zero or deterministic rows may use fewer bits.

The net and an added rational unit seed are explicit. For every command-vertex pair the exact enumeration in Lemma 9.1 examines at most $\binom{k-1}{3}$ triples, where $k \leq 13m^2 + 1$. The total is $O(|A|k^4)$ constant-size rational linear-algebra operations, in addition to the probe tables. All determinants, comparisons and row identities involve rational numbers with polynomially bounded

bit lengths in the supplied data and $\log m$; dyadic rounding adds b bits. The computation of m and b can use integer comparisons, not floating-point decisions. This proves the stated uniform, numerical-parameter complexity. It is a construction, not a minimum-width algorithm, and its exhaustive fallback need not be practical for large horizons. $\square$

11.1. An effective process converse for the rational free experiment. Keep the rational rotations and seed of (8.2)–(8.3), now exposing the three nondisturbing binary probes

$$(11.6) \quad q_{j,+}(x) = (1 + \lambda x_j)/2, \quad q_{j,-}(x) = (1 - \lambda x_j)/2, \quad j = 1, 2, 3,$$

where $0 < \lambda < 1$ is rational. This defines a fixed rational classical experiment. The probing law is not an implementation of nondisturbing quantum measurement. Let PW refer to the causal model above, not to terminal density-matrix width W .

Theorem 11.2 (Effective repeated-observation lower bound). *Let $0 \leq \varepsilon < 1$ and $0 < \delta < 1 - \varepsilon$. For an integer $t \geq 0$ put*

$$(11.7) \quad L_t = \max \left(1, \left\lceil 18\lambda^{-2}625^t \log(6/\delta) \right\rceil \right).$$

At every horizon $N \geq t + 3L_t$, any error- ε causal machine has

$$(11.8) \quad |S_t| \geq (1 - \varepsilon - \delta)(2 \cdot 3^t - 1).$$

If $\varepsilon + \delta < 1/2$, it has the stronger lower bound $|S_t| \geq 2 \cdot 3^t - 1$. More generally, the same bounds hold at every cut u with $t \leq u \leq N - 3L_t$ after padding the prefixes by probes. In particular, for each fixed $0 < \varepsilon < 1$, there are explicit positive constants $c_{\varepsilon,\lambda}, C_{\varepsilon,\lambda}$ such that, for all sufficiently large N ,

$$(11.9) \quad c_{\varepsilon,\lambda} N^{\log 3 / \log 625} \leq \text{PW}_{N,\varepsilon} \leq C_{\varepsilon,\lambda} (N + 1)^2.$$

The converse uses no spectral-gap theorem. Its exponent is not asserted to be sharp.

Proof. The rational freeness and stabilizer argument in Theorem 8.2 gives $2 \cdot 3^t - 1$ different reachable pure Bloch vectors at time t . Their coordinates lie in $25^{-t}\mathbb{Z}^3$. Hence each distinct pair differs in some coordinate by at least 25^{-t} , and the associated plus-probabilities in (11.6) differ by at least $\lambda 25^{-t}/2$. Use one common continuation: repeat each of the three probes L_t times. Hoeffding's elementary Bernoulli bound and a union bound give

$$\mathbb{P}_x \left\{ \max_j |\hat{q}_j - q_{j,+}(x)| \geq \lambda 25^{-t}/6 \right\} \leq 6 \exp(-L_t \lambda^2 625^{-t}/18) \leq \delta.$$

The empirical neighborhoods for the different targets are disjoint. At a cut label z , let Q_z be the law of this same continuation. Marginalizing prefix outputs and using total-variation correctness shows that the continuation law from each prefix gives its own neighborhood probability at least $1 - \varepsilon - \delta$. Writing this law as $\sum_z \alpha_x(z) Q_z$ and summing over the disjoint neighborhoods yields (11.8). If the probability exceeds $1/2$, each neighborhood needs a different label assigning it probability greater than $1/2$, giving the stronger conclusion. The proof does not condition on the success of a previous observation. Padding by arbitrary fixed probes leaves the target states unchanged and proves the assertion at later cuts. Unused final controls may also be probes. A terminal cut without enough remaining observations is not claimed to satisfy this lower bound.

For the power lower bound take $\delta = (1 - \varepsilon)/2$ and $B = 54\lambda^{-2} \log(6/\delta) + 4$. Then $t + 3L_t \leq B625^t$. For $N \geq B$, choose $t = \lfloor \log_{625}(N/B) \rfloor$. Since $3^t \geq (N/B)^{\log 3 / \log 625}/3$ and $2 \cdot 3^t - 1 \geq 3^t$, (11.8) gives, for example, $c_{\varepsilon,\lambda} = (1 - \varepsilon)B^{-\log 3 / \log 625}/6$. The upper bound follows from Theorem 11.1 with $\eta = (1 - \lambda)/2$ and $A = \lambda^2/2$. For irrational requested ε , construct at any rational tolerance between $\varepsilon/2$ and ε . All constants can thus be bounded from the displayed data without computing a group spectral gap. $\square$

The tests in this theorem consume horizon. The full-state terminal exponential profile and the causal lower bound therefore express different requirements. Conversely, the causal upper construction controls one joint transcript law under adaptive control, not just its one-time marginals. These process statements do not by themselves sharpen the terminal spectral-gap converse. Its logarithmic loss is removed separately by Theorem 4.5.

12. FINITE PRECISION FOR SUPPLIED REALIZATIONS

Lemma 12.1 (Dyadic rounding of a probability row). *A probability row p with m entries has a dyadic row $\hat{p}$ of precision b such that*

$$\text{TV}(p, \hat{p}) \leq (m-1)2^{-b}.$$

One exact draw from $\hat{p}$ uses at most b independent fair bits.

Proof. Round the first $m-1$ entries down to multiples of 2^{-b} and put the remaining mass in the last entry. This is a probability row. Its total variation error equals the mass transferred to the last entry, which is at most $(m-1)2^{-b}$. For sampling, draw a uniform integer in $\{0, \dots, 2^b-1\}$ using b bits and compare it with the integer cumulative row sums. Degenerate rows can use fewer bits. $\square$

13. POSITIVE FINITE-PRECISION MATRIX DECODERS

The label-width bounds leave real table descriptions uncharged. There is nevertheless a finite-description counterpart for each supplied encoded realization. Entrywise rounding of a density matrix is unsafe: it need not preserve positivity. A Gram-factor construction avoids this problem.

Lemma 13.1 (Rational positive rounding). *Let $D \in \mathcal{D}_d$ and let $L = D^{1/2}$. Approximate the real and imaginary parts of every entry of L by dyadic multiples of 2^{-b} , each within 2^{-b} , to obtain Q . Set $\delta = \sqrt{2}d2^{-b}$. If $\delta \leq 1/4$, then*

$$(13.1) \quad \hat{D} = \frac{QQ^*}{\text{tr}(QQ^*)} \in \mathcal{D}_d \cap M_d(\mathbb{Q} + i\mathbb{Q}), \quad \|\hat{D} - D\|_F \leq 8\sqrt{2}d2^{-b}.$$

The construction is effective when D is given by exact algebraic entries. Its rational output has $O(d^2(b + \log(d+1)))$ bits for fixed encoding conventions.

Proof. Write $Q = L + E$. Then $\|E\|_F \leq \delta$ and $\|L\|_F^2 = \text{tr } D = 1$. Hence $Q \neq 0$, $\text{tr}(QQ^*) \geq (1-\delta)^2$, and

$$\|QQ^* - D\|_F \leq (2+\delta)\delta, \quad |\text{tr}(QQ^*) - 1| \leq (2+\delta)\delta.$$

The first estimate follows from the two cross terms and EE^* , and the second from $\|Q\|_F^2 - \|L\|_F^2$. Since $\|D\|_F \leq 1$, division by $\text{tr}(QQ^*)$ gives an error at most $2(2+\delta)\delta/(1-\delta)^2 \leq 8\delta$. Positivity and trace one are exact, not numerical tolerances. After multiplying Q by 2^b , the normalization is a ratio of integer Gram entries to a positive integer sum of squares. Their bit sizes are $O(b + \log(d+1))$. The positive square root of an algebraic positive matrix has algebraic entries and can be specified and approximated by real-algebraic arithmetic. This proves the effectivity assertion without an efficiency claim for obtaining D . $\square$

Theorem 13.2 (A supplied realization with finite tables and fair bits). *Suppose an error- ε horizon- N realization for a matrix interface uses at most k labels, and its initialization, command rows, and decoders are supplied with exact algebraic encodings. Let $\sqrt{2}d2^{-b} \leq 1/4$. There is a rational finite-table implementation, with the same persistent label widths, whose Frobenius error is at most*

$$(13.2) \quad \varepsilon + \sqrt{2}((N+1)(k-1) + 8d)2^{-b}.$$

Each hidden draw uses b independent fair bits, so at most $(N+1)b$ such bits suffice. For a supplied stationary permutation realization the additional error is at most

$$(13.3) \quad \sqrt{2}(k-1+8d)2^{-b},$$

and only b fair bits are needed for initialization, independently of the horizon. The numerical matrix decoder itself is not a quantum preparation or a sampled terminal answer register.

Proof. Apply Lemma 12.1 to each initialization and command row after padding to k labels. Each row changes by at most $(k-1)2^{-b}$ in total variation. Induction, or a maximal coupling at every actual draw, bounds the terminal distribution discrepancy by $(N+1)(k-1)2^{-b}$; this upper bound may harmlessly exceed one. Replace each decoder by the legal rational matrix in Lemma 13.1. The Frobenius diameter of $\mathcal{D}_d$ is $\sqrt{2}$, so changing the distribution incurs at most $\sqrt{2}$ times its total-variation discrepancy. The independent decoder error is at most $8\sqrt{2}d2^{-b}$. This proves (13.2). A dyadic probability row of denominator 2^b is sampled by one uniform b -bit integer and its cumulative integer table.

For a permutation realization the command rows already have exact integer entries and need not be rounded or sampled. Only the initialization distribution and decoder are changed. The same estimate therefore gives (13.3) and the horizon-independent fair-bit bound. $\square$

The row tables and their cumulative sums require $O((N|\mathcal{A}|k^2 + |\mathcal{S}|k)b)$ bits in the general case, in addition to $O(kd^2(b + \log(d+1)))$ bits for one matrix decoder. Temporary random integers and table indices require their own $O(b + \log(k+1))$ workspace; this workspace is not hidden in the persistent-label invariant. Multiple queries multiply the decoder storage by their number. To implement a sampled measurement one must additionally specify and charge its sampler. The theorem compiles a supplied encoded realization; it neither computes a width minimizer nor assigns finite encodings to arbitrary unspecified real tables. Applying it after the exact inner-polytope construction uses an explicit extra error budget, not an exact finite-precision claim at an extreme zero-error endpoint.

Here an exact algebraic entry is specified by defining polynomials and real isolating intervals (for both parts of a complex entry), with compatible field operations. The positive square root is the unique positive semidefinite solution of $L^2 = D$, and its isolating data can be obtained by real-algebraic algorithms. These representation and output bit-length statements are not running-time bounds for finding the original realization or its square root. The constructed rational qubit compiler is a separate, more explicit algorithm.

14. RELATION TO REALIZATION AND AUTOMATA THEORY

The polynomial matrix upper construction realizes an entire legal terminal density matrix, not merely one acceptance probability. In particular, enlarging its decoder set to the full probability simplex after tomography changes the problem. The extreme-point proof requires the legal image of $\mathcal{D}_d$.

Brodsky–Pippenger [9, Theorems 3.3 and 3.6; Lemma 3.5] characterize bounded-error measurement quantum language recognition by group languages, discuss a free-group word problem, and transfer cut-point recognition to probabilistic automata. Those statements compare languages and cut-points; our exact profile constrains the full terminal output and every cut of a horizon-specific stochastic machine. Ambainis–Freivalds [10] likewise provide an important automata-complexity comparison, not a substitute for that legal-output quantifier.

The geometric ingredients are classical smooth homogeneous-space and polytope arguments; compare [14, 15]. The least-rank theorem makes the cap estimate uniform over all conditional-centroid directions, including singular orbit types. The quaternion argument gives complete finite-input freeness and stabilizer proofs. Bourgain–Gamburd [16, Theorem 1] is a separate

deep input for the actual displayed alphabet’s noisy converse. Neither that spectral theorem nor classical free subgroup constructions are counted as new realization theorems.

Positive realization and invariant-cone methods [3, 7] explain why a common positive lift, rather than unrelated one-cut factorizations, is necessary. Hidden-process realization [4] concerns consistent output processes; the terminal model is a controlled convex-valued interface. The separate adaptive construction below specifies a nonnegative emission instrument and a whole-transcript error bound, rather than inferring a process law from its marginals. These comparisons specify different mathematical questions, not an exhaustive priority determination.

14.1. Entropy transport and the sharp converse. Polyanskiy–Wu [27, Proposition 1] prove Wasserstein continuity of entropy for Euclidean densities under pointwise logarithmic-gradient regularity. The entropy–transport principle itself is not new. Our spherical mixture estimate is proved directly because compactly supported kernels vanish and do not satisfy that regularity assumption. The additional realization steps are the transport coupling paid for by *centroid mass* loss, the entropy defect on unions of orbitwise small caps, and telescoping along all narrow cuts with no restriction on intermediate widths. These give the lower power without a repeated mixing-block logarithm.

The Hecke spectral bound [28, 29] is classical and deep. It supplies the numerical norm of the actual six-rotation operator, not a numerical experiment or a newly proved spectral theorem. The contribution here is its combination with the stochastic occupation argument, explicit coset profile and sparse rational upper machine. The first rational-unitary example still imports a non-effective Bourgain–Gamburd constant; the second rational Bloch example has the displayed numerical width constants.

The logarithmic mismatch in the fixed-error law is thereby removed when least-orbit and target-orbit dimensions agree. Distinct dimensions can still give different lower and upper powers. We do not determine optimal leading constants, the full error–horizon crossover, or rates for every compact representation or alphabet. The stated finite-table algorithms are polynomial in numerical horizon and inverse accuracy, not in their binary lengths. General compact-group minimization has no practical total-complexity claim here.

14.2. A distinct causal approximation problem. The process compiler constructs a non-negative emission instrument, not a collection of separately accurate matrix decoders. It uses a supplied rational unit seed, rational rotations and a positive probability floor for affine probes. The entropy comparison is performed on an auxiliary joint trajectory with the target public marginal, so adaptive actions are covered without conditional-error claims after rare histories. The process algorithm is not the proof of the sharp terminal law. That law uses the separate spectral-entropy argument and its own assumptions; the effective spherical instance uses the LPS bound. The elementary process converse uses instead finite statistical tests and the rational orbit-separation lattice.

REFERENCES

Stochastic semigroups and positive realization

- [1] P. Flor, On groups of non-negative matrices, *Compositio Mathematica* **21** (1969), no. 4, 376–382.
 - [2] Š. Schwarz, On the structure of the semigroup of stochastic matrices, *Publications of the Mathematical Institute of the Hungarian Academy of Sciences, Series A* **9** (1964), 297–311.
 - [3] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Transactions on Automatic Control* **49** (2004), no. 5, 651–664.
 - [4] A. Heller, On stochastic processes derived from Markov chains, *Annals of Mathematical Statistics* **36** (1965), no. 4, 1286–1291.
- ### *Automata, weighted series, and recognition*
- [5] M. O. Rabin, Probabilistic automata, *Information and Control* **6** (1963), no. 3, 230–245.
 - [6] A. Paz, *Introduction to Probabilistic Automata*, Academic Press, New York, 1971.

- [7] J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Encyclopedia of Mathematics and its Applications 137, Cambridge University Press, Cambridge, 2011.
- [8] N. Fijalkow and C. Paperman, Monadic second-order logic with arbitrary monadic predicates, arXiv:1709.03117 (2017).
- [9] A. Brodsky and N. Pippenger, Characterizations of 1-way quantum finite automata, *SIAM Journal on Computing* **31** (2002), no. 5, 1456–1478.
- [10] A. Ambainis and R. Freivalds, 1-way quantum finite automata: strengths, weaknesses and generalizations, arXiv:quant-ph/9802062 (1998).
- [11] Y. Feng, L. Song, and L. Zhang, Distribution-based bisimulation and bisimulation metric in probabilistic automata, arXiv:1512.05027 (2015).
- [12] B. Steinberg, Topological dynamics and recognition of languages, arXiv:1306.1468 (2013).
Compact groups and orbit geometry
- [13] G. B. Folland, *A Course in Abstract Harmonic Analysis*, second ed., Chapman & Hall/CRC, Boca Raton, 2016.
- [14] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, second ed., Graduate Texts in Mathematics 222, Springer, Cham, 2015.
- [15] S. J. Szarek, Metric entropy of homogeneous spaces, *Banach Center Publications* **43** (1998), 395–410.
- [16] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *Journal of the European Mathematical Society* **14** (2012), no. 5, 1455–1511.
- [17] E. Breuillard and T. Gelander, On dense free subgroups of Lie groups, *Journal of Algebra* **261** (2003), no. 2, 448–467.
- [18] E. Breuillard and T. Gelander, On dense free subgroups of Lie groups—revisited, arXiv:2605.20568 (2026).
Algebraic computation and complexity
- [19] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second ed., Algorithms and Computation in Mathematics 10, Springer, Berlin, 2006.
- [20] V. D. Blondel, E. Jeandel, P. Koiran, and N. Portier, Decidable and undecidable problems about quantum automata, *SIAM Journal on Computing* **34** (2005), no. 6, 1464–1473.
- [21] H. Derksen, E. Jeandel, and P. Koiran, Quantum automata and algebraic groups, *Journal of Symbolic Computation* **39** (2005), nos. 3–4, 357–371.
- [22] W. A. de Graaf, Computing the Zariski closure of a finitely generated matrix group, arXiv:2505.02577v2 (2026).
- [23] K. Nosan, A. Pouly, S. Schmitz, M. Shirmohammadi, and J. Worrell, On the computation of the Zariski closure of finitely generated groups of matrices, arXiv:2106.01853v6 (2025).
- [24] Y. Shitov, A universality theorem for nonnegative matrix factorizations, arXiv:1606.09068v2 (2018).
- [25] F. P. Ramsey, On a problem of formal logic, *Proceedings of the London Mathematical Society* (2) **30** (1930), no. 1, 264–286.
- [26] S. Singh, M. R. James, and M. Rudary, *Predictive state representations: A new theory for modeling dynamical systems*, Proceedings of the Twentieth Conference on Uncertainty in Artificial Intelligence, AUAI Press, 2004, 512–519. See Theorems 2 and 6; arXiv:1207.4167.
Entropy transport and explicit spherical spectral bounds
- [27] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Transactions on Information Theory* **62** (2016), no. 7, 3992–4002. DOI: 10.1109/TIT.2016.2562630.
- [28] A. Lubotzky, R. Phillips, and P. Sarnak, Hecke operators and distributing points on the sphere I, *Communications on Pure and Applied Mathematics* **39** (1986), suppl., S149–S186. DOI: 10.1002/cpa.3160390710.
- [29] A. Pinochet Lobos and C. Pittet, The exact convergence rate in the ergodic theorem of Lubotzky–Phillips–Sarnak, arXiv:1805.05261v2 (2019), Theorem 1.1. Published in expanded form as *The exact convergence rate in the ergodic theorem of Lubotzky–Phillips–Sarnak and a universal lower bound on discrepancies*, *L’Enseignement Mathématique* **67** (2021), nos. 1–2, 63–94. DOI: 10.4171/LEM/1003.